Science
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Supplementary Materials for

Targeted MYC2 stabilization confers citrus Huanglongbing resistance

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Science **388**, 191 (2025)
DOI: 10.1126/science.adq7203

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MDAR Reproducibility Checklist
Data S1 to S5

Materials and Methods

Plant materials and growth conditions

The citrus trees used in this study were mainly Hongjiang orange (*Citrus sinensis* L.) except specifically mentioned elsewhere. All citrus species and other Rutaceae plants were collected and grown in the greenhouse in Beijing, China (IMCAS) with a growth temperature of 27°C, 65% relative humidity with a 16-hour light/8-hour dark cycle. *Nicotiana benthamiana* plants grew in an insect-free growth chamber at 25°C with a 12-hour light/12-hour dark cycle.

Plasmid constructs

For transient expression construction, the coding region of *SDEs* without the signal peptide was separately amplified from *C*Las genomic DNA using gene-specific primers with a start codon added to the 5′ end of the *SDEs* forward primer. The coding regions of PUB21, PUB21DN, SDE5, and MYC2 were cloned into pH7WGY2 or pBA-myc to generate YFP or myc-tag fusion vectors. The 600bp and 800bp coding regions of PUB21-specific fragments were cloned into the *pH7GWIWG2(II)* vector to generate *PUB21* RNAi-1 and *PUB21* RNAi-2 vectors. For virus-induced gene silencing (VIGS) vectors, the 800bp coding region of the MYC2-specific fragment was cloned into the *psTRV2* vector to generate the *psTRV2-MYC2* vector.

For recombinant proteins, full-length open reading frames encoding *PUB21* or *MYC2* were cloned into pGEX-DC or pMAL-DC to generate GST or MBP tag fusion construct. The 336 bp coding region of *SDE5* without the signal peptide (1-20 aa) was separately cloned into the pET-28a vector to generate the His-SDE5 fusion construct. These protein expression constructs were transformed into *Escherichia coli* BL21(DE3). Recombinant proteins were purified following the manufacturer's instructions.

For bimolecular fluorescence complementation (BiFC), the DNA fragments of *SDEs, MYC2*, *PUB21* and its deletion derivatives were cloned into the pENTR-3C entry vector. And then transferred into pBA3032 and pBA3036 destination vectors by LR recombination reactions to generate fusion genes with cEYFP or nEYFP at the amino- or carboxy-terminus, under the control of the *CaMV 35S* promoter. Constructs for BiFC were transferred into *A. tumefaciens* strain EHA105.

To enhance the editing efficiency of the prime editing system (PE), we optimized a rice PE vector specifically for citrus PE by using CsU6 and CmYLCV promoters (*22*, *23*). Subsequently, we used the PlantPegDesigner website to design peg-PUB21 primer containing a 10-nt primer binding site (PBS) and a 14-nt RT template for the substitution of TGC for GCC at the amino acid 39 of PUB21. Then the peg-PUB21 was amplified using a forward primer containing the target site and a reverse primer containing the PBR+RT sequence with sgRNA as templates, and then inserted into PE2 (Cs) using the Gibson assembly method. The constructs were transformed into *Escherichia coli* DH5α for amplification.

Antibody preparation

The DNA fragment of SDE5 or PUB21 was cloned into pET-28a (+) vector to generate His-tagged constructs. His-tagged protein was purified using Ni-NTA agarose (Qiagen, R90101) according to the manufacturer's instructions. His-tagged protein was then injected into rabbits, and the corresponding polyclonal antibody was prepared by the Beijing Protein Innovation Co., Ltd.

Citrus protoplast transfection

The *C. sinensis* leaves were used to prepare protoplasts. Protoplast isolation and transfection were performed as described previously (*24*). For each construct, 10 μg of plasmid was delivered into protoplast cells by polyethylene glycol-mediated transfection, and then the treated protoplasts were incubated at 23°C in the dark. After a 48-h incubation, genomic DNA was extracted using the CTAB method for amplicon sequencing.

sTRV-based VIGS in citrus plants

The synthetic tobacco rattle virus (sTRV) system-based VIGS was performed as described previously (*25*). *C. sinensis* seedlings with two true leaves were *Agrobacterium*-mediated vacuum infiltration with *psTRV1* and *psTRV2-MYC2* to generate *MYC2* VIGS plants. Plants co-infiltrated with *psTRV1* and *psTRV2* were used as vector controls.

Generation of transgenic citrus

For citrus genetic transformation, the full-length cDNA of *PUB21, PUB21DN* and *SDE5* without the signal peptide was inserted into a pLGN135 vector that was driven by the *35S* promoter (*26*). The constructs were respectively introduced into the *A. tumefaciens* strain EHA105. Epicotyl segments from *C. sinensis* seedlings were used for *A.tumefaciens*-mediated transformation as previously described (*27*). Transgenic shoots were then identified by GUS staining as well as PCR analysis before being micrografted onto *Citrus aurantium* seedlings. Subsequently, the resulting plantlets were further grafted onto *Citrus aurantium* seedlings in a glasshouse. The expression of *PUB21, PUB21DN* and the *SDE5* effector in those transgenic plants was determined by RT-qPCR analysis.

***C*Las challenge on citrus trees, detection and quantification**

For psyllid-mediated *C*Las transmission, leaves of 1-year-old healthy plants per independent transgenic line (including healthy vector control) were dipped and fixed in liquid MS solution for feeding ten *C*Las-infected *D. citri* adults and treated with plastic cups in airtight confinement for 7 days, with leaves of comparable size and color. The *D. citri* adults were tested in advance by PCR to determine the carrying of *C*Las. After 7 days, the *C*Las-infected *D. citri* adults were removed and the leaves were collected. Each treatment was replicated 3 times.

For *C*Las transmission using the grafting method, one-year-old healthy individuals from two transgenic lines with comparable height and crown were used to infect HLB. Before grafting, PCR was used to detect the presence of *C*Las in the petioles of the leaves. Transgenic lines were used to infect *C*Las by grafting HLB-positive excised branches to them. To isolate DNA, three randomly selected midribs from each plant were combined into a single sample at regular intervals. The disease intensity for each independent line was estimated based on *C*Las populations from three grafted transgenic trees.

The midvein of leaves was dissected from *C. sinensis* leaves. The DNA extracts were obtained by a standard CTAB method. Quantification of *C*Las in plant tissues was performed as previously described (*28*). The qPCR reactions were carried out with HLBas and HLBr primers for *C*Las and one internal control primer COX for citrus DNA. Quantification of *C*Las was conducted using the following equation: Y=–0.2898X+10.143. The *C*Las also was tested by PCR with specific primers A2/J5 (*29*).

SDE5-based detection of the *C*Las method

Protein extraction from the phloem sap was performed following previous reports (*30, 31*). Phloem-rich tissues (bark and midribs) were excised from *C*Las-infected and uninfected citrus trees. These tissues were rinsed with deionized water and then cut into about 1 cm pieces using a

sterile razor. These pieces were placed in a 2mL separation column and centrifuged for 20 min at 15,000 g at 4°C. The collected phloem sap was heated for 5 min at 95°C in the SDS gel-loading buffer. Western blotting was performed for SDE5 detection with the specific antibody. Tissues from healthy citrus trees were used as the control.

For the citrus immunological imprinting assay, *C*Las status was verified by RT-PCR with A2/J5 primers before imprinting. Freshly cut phloem-rich tissues (bark and midribs) of *C*Las-infected and uninfected citrus were imprinted by gently pressing on nitrocellulose membranes. The printed membranes were dried overnight at 4°C, then washed in 1×TBST buffer (25 mM Tris-HCl pH7.5, 150 mM NaCl, 0.1% Tween-20) at room temperature for 30 min. Membranes were blocked with 1×TBST containing 5% w/v non-fat milk at room temperature for 3 h and then incubated with the SDE5-specific antibody at room temperature for 2 h. The membranes were washed three times with 1×TBST (10 min each time) and incubated at room temperature for 1 h with the secondary antibody, AP-conjugated goat anti-rabbit IgG (H+L). Signals were detected by the BCIP/NBT Chromogenic Kit (PR1100-125ml, Solarbio Company, China) following the manufacturer's instructions.

Transposon prediction

To validate the potential transposon characteristics of the 150-bp insertion, a transposable element library of the *Citrus sinensis* genome was generated using EDTA (*32*). Subsequently, we performed sequence alignment of the above sequence against the transposon library using Repeatmasker (v4.1.1), with the thresholds and parameters set to the default values.

E3 ligase gene family in different citrus organisms

For protein sequence annotation, due to the lack of annotation information for the E3 ligase gene family in the citrus reference genome and annotation files of different citrus organisms, we performed protein domain annotation with the following steps to identify genes belonging to the E3 ligase gene family. We used the "interproscan.sh" script from the InterPro database (version: 5.25-64.0) (*33*) to perform domain annotation on the protein sequence files of different citrus organisms. The parameters used were "-f tsv -goterms -i -appl Pfam -iprlookup -o". The Pfam annotation, InterPro annotation, and Gene Ontology (GO) information were used for further analysis.

For protein sequence alignments, we created a reference database using 1,515 E3 ligase protein sequences from rice, as identified in previous studies (*34*). The reference database was constructed with the parameters "makeblastdb -in -dbtype prot". Next, BLAST was used to align the protein sequences of different citrus organisms with the reference database. The parameters used were: "blastp -query -db -out -outfmt "6 qaccver saccver pident length mismatch ppos gapopen qstart qend sstart send evalue bitscore qlen slen ssciname sskingdom staxid stitle" -max_target_seqs 1 -max_hsps 1 -seg no". Finally, we integrated the differential expression analysis results, protein domain annotation results, and protein sequence alignment results of different citrus organisms for further analysis of the E3 ligase gene family.

Prediction of antimicrobial peptides (AMPs)

Small open reading frames (sORFs) of the human gut microbiome were collected from the DBsmORF (http://104.154.134.205:3838/DBsmORF/) database (*35*). After removing redundant sequences and sequences containing one or more non-standard amino acids, 122826 sORFs were left. SORFs were annotated using SmORFinder default parameters. AMPDiscover was used to

identify AMPs from sORFs (*36*). Representative AMPs were chosen by using MMseqs2, with an amino acid identity of 50% and coverage of 80%.

Peptide-protein molecular docking

The structures of PUB21 and peptides were predicted using Modeller. The modelled structure of PUB21-peptide complexes was based on the docking by Autodock vina. The structure data were processed using the program PyMOL software.

Quantitative RT-PCR

Total RNA was isolated from citrus plants using the RNeasy plant mini kit (Qiagen, 74904). Reverse transcription was performed using 2μg RNA of each sample and TransScript One-Step gDNA Removal and cDNA Synthesis SuperMix (TransGen Biotech, China). Three independent biological samples were collected and analyzed. Real-time PCR was performed on a Bio-Rad CFX96 real-time PCR system with ThunderbirdTM SYBR qPCR Mix (TOYOBO, QPS-201) and gene-specific primers listed in table S6. The citrus COX gene was used as an internal control.

Ubiquitination assays

The *in vitro* ubiquitination assays were performed as described previously (*9, 37*). The 30 μL reaction mixture contained 200 ng AtUBA1 (E1), 200 ng AtUBC8 (E2), 2 μg ubiquitin (Sigma-Aldrich, U5382), 500 ng PUB21-myc, PUB21DN-myc, and with or without 500 ng MBP-MYC2. GST-SDE5 was added to the reactions following indicated times. Reactions were incubated at 30°C for 2 h. For ubiquitination assays in *E. coli*, the coding region of *MYC2*, *PUB21*, or *PUB21DN* genes was PCR amplified from *C. sinensis* cDNA and cloned into the pCDFDuet or pACYCDuet vectors. *SDE5* was cloned into pGEX-DC to generate a GST-SDE5 fusion construct. *E. coli* strain BL21 (DE3) competent cells containing different combinations of the expression vectors were cultured and then induced by the addition of 0.5 mM isopropyl beta-D-1-thiogalactopyranoside (IPTG, Merk, V900917). For screening the inhibitors for PUB21 E3-ligase activity, the peptide candidates were added to the inducted bacterial cells. After induction, bacterial cells were grown at 25 ˚C for 10 h and then stored at 4˚C overnight. The crude protein extracts were separated by 6% or 8% SDS–PAGE gels and analyzed by immunoblotting using an anti-MBP antibody or anti-myc antibody.

MG132 and Cycloheximide Treatments

Transgenic citrus leaves were injected with 100 μM MG132 (Sigma, M8699) for 16 h. To examine the half-life of MYC2, transgenic citrus leaves treated with MG132 as above were washed five times before being treated with 200 μM cycloheximide (Sigma, C7698) to block de novo protein synthesis. Treated samples were collected at the indicated time points for immunoblot analysis.

Yeast two-hybrid analysis

All constructs were transformed into yeast strain AH109 through the method of modified lithium acetate. AD-MYC2 or BD-PUB21 was then used to screen against the full yeast library using a mating-based yeast-two-hybrid (Y2H) approach. To confirm the interaction between SDEs and MYC2 or SDE5 and PUB21, the full-length citrus MYC2 was cloned into the pGAD424 vector through LR reaction to generate AD-MYC2. The yeast strain AH109 was co-transformed with AD-fusions construct and BD-fusion constructs. Yeast cotransformed with the indicated plasmids was spotted onto a synthetic medium (SD-Leu-Trp-His) containing 5 mM 3-amino-1,2,4-triazole. The empty vectors pGBKT7 (BD) and pGADT7 (AD) were used as negative controls.

Bimolecular fluorescence complementation (BiFC)

Fluorescence was observed owing to the complementation of the SDEs or PUB21-fused with the C-terminal part of EYFP and MYC2-fused with the N-terminal part of EYFP. The nEYFP-GUS and cEYFP-GUS were used as negative controls. *Agrobacterial* EHA105 cells containing the indicated constructs were infiltrated into the leaves of 4-week-old *N. benthamiana* plants. After infiltration for 2-3 days, fluorescence signals of EYFP were recorded by SP8 confocal microscopy.

In vitro pull-down and protein competitive interaction assay

The GST- and His-tagged proteins were separately purified using Glutathione sepharose (GE Healthcare, 17-5132-01) and Ni-NTA agarose (Qiagen, R90101) beads according to the manufacturer's instructions. For *in vitro* binding assay, GST-MYC2 or GST-PUB21 and His-SDE5 fusion proteins were mixed and incubated with 50 μL GST-Trap for 2 h at 4˚C in 1mL binding buffer (50 mM Tris-HCl, pH 7.5, 100 mM NaCl, 0.5% Triton X-100, and 35 mM b-mercaptoethanol). The pulled-down proteins were resuspended in SDS gel-loading buffer and detected by immunoblots using anti-GST (TransGen Biotech, HT601-02) and anti-His (TransGen Biotech, HT501-02) antibodies.

For competitive pull-down assay, indicated amounts of His-SDE5 or MBP were mixed in 1mL binding buffer with 2 μg of GST-PUB21 fusion proteins and 50 μL of Amylose resin beads at 4˚C overnight. After four washes, 2 μg of MBP-MYC2 fusion proteins were added and the mixture was incubated for 3 h at 4˚C. After incubation, binding beads were washed 6 times. The associated proteins were separated on 10% SDS-polyacrylamide gels and detected by immunoblots using an anti-MBP antibody.

ChIP assays

Healthy plant DNA was extracted by the CATB method. Purified MBP or MBP-MYC2 proteins were incubated with citrus DNA. Anti-MYC2 resin beads were incubated with DNA from *M. paniculata* and *B. koenigii*. ChIP experiments were performed as previously described for co-immunoprecipitation (*11*). The resulting DNA samples were purified with the PCR purification kit. DNA fragments were analyzed by quantitative PCR. The fragment of citrus COX promoter was used as a reference. Primers of ChIP assays are listed in table S6.

Electrophoretic mobility shift assays (EMSA)

EMSA was performed as described previously (*11*). MBP-MYC2 or MBP proteins were purified from *Escherichia coli* BL21 cells using amylose resin beads (New England Biolabs, E8021S). The fragments containing G-box like *cis*-elements (CANNTG) or mutant G-box like *cis*-elements (TTNNAA) in the *PUB21* promoters were labeled by Biotin (table S6). EMSA was performed using a LightShift Chemiluminescent EMSA Kit (20148, Thermo Scientific) according to the manufacturer's instructions.

Microscale thermophoresis (MST) assay

The affinities between BSA, APP3, APP3-14 and GST-PUB21 were measured with an MST assay in a Monolith NT.115 instrument (NanoTemper Technologies). GST-PUB21 was labeled with the blue fluorescent dye NHS-495. The concentration of GST-PUB21 was maintained at 10 μM, whereas the concentrations of APP peptide were gradient diluted from 500 μM to 15.26 nM. After a brief incubation, MST standard glass capillaries (NanoTemper Technologies) were filled with the samples. The measurements were performed at 25°C. The assays were repeated in three

biological replicates. Data were analyzed to obtain KD using Monolith Affinity Analysis version 2.2.4 software.

Luciferase activity assay

The different *PUB21* promoter: luciferase was used as a reporter construct. The *35S:YFP* and *35S:MYC2* were used as effector constructs. The *N. benthamiana* leaves were co-infiltrated with the indicated constructs. After incubation for 2 days, samples were collected and the luciferase (LUC) activity was determined by a microplate reader as described (*38*). Eight biological samples were used in one experiment.

GUS staining and GUS activity assay

Leaves of *N. benthamiana* were co-infiltrated with different *PUB21* promoter: *GUS* and 35*S:YFP* or 35*S:YFP-MYC2* at a ratio of 1:1. Two days after infiltration, leaves were harvested and stained with GUS staining buffer [0.1 M sodium phosphate buffer (pH 7.0), 0.1% Triton X-100, 0.5 mM $K_3Fe(CN)_6$, 0.5 mM $K_4Fe(CN)_6$, and 10 mM EDTA]. Leaves were collected and frozen in liquid nitrogen, and a GUS quantitative assay was performed following the previously described procedure (*38*). Eight biological samples were used for each experiment.

Detection of MYC2 protein

Protein extraction from Aurantioideae species leaves was resolved in an 8% SDS-polyacrylamide gel electrophoresis (PAGE) and transferred onto a polyvinylidene fluoride membrane. Immunoblot analysis was performed with an anti-MYC2 polyclonal antibody (AB clonal, A20774) and an anti-rabbit horseradish peroxidase-linked sheep antibody.

Jasmonate (JA) and its derivatives analysis

Leaf of 150 mg was extracted with 1 mL of ethyl acetate containing 10 ng of D6-JA (ISOREAG, IR-59670) (*39*). All samples were then vortexed for 10 min and centrifuged at 14,000 rpm for 20 min at 4 °C. Supernatants were collected and evaporated using a vacuum concentrator to dry at 30 °C. Residues were resuspended in 500 μL of MeOH: H_2O (70:30, v/v) and centrifuged at 14,000 rpm for 10 min. The supernatants were then collected and analyzed with a high-performance liquid chromatography-tandem mass spectrometry system (QTRAP 4500, AB Sciex). The MS was operated in negative mode and Multiple Reaction Monitoring (MRM) of ion pairs was used for the analysis of D6-JA (215.0>59), endogenous JA (209.2>59) and endogenous JA-Ile (322.1>130). Experiments were performed with three independent biological replicates.

Syringe injection and trunk injection of diseased citrus trees

To quickly evaluate the ability of APPs to treat HLB diseases, APPs solution was injected into live leaves of CLas-infected 3-year-old Hongjiang orange trees using a syringe. After treatment for 48 h, 6 treated leaves in 3 branches were collected and analyzed *C*Las titers. Antibiotic tetracycline was used for comparative analysis. In greenhouse experiments, approximately 1 μM BSA or 1 μM APP3-14 peptides at 300 ml water were separately injected into the trunk of *C*Las-infected two 3-year-old trees and four 1-year-old trees. In field trials, approximately 1 μM BSA or 1 μM APP3-14 peptides at 500 ml water were separately injected into the trunk of *C*Las-infected four 3-year-old orange trees. HLB disease incidence in different treatments was evaluated by disease symptoms and the bacterial load of *C*Las. *C*Las titers in different treatments were investigated by evaluating 3 branches in each treated tree.

Transmission electron microscopy

*C*Las-infected citrus leaves were incubated with 10 μM Tet, 1 μM APP3 and APP3-14 for 48 h in a growth chamber. Leaves incubated with water for 48 h were used as mock treatment. The midvein of the leaves was dissected. The tissues were fixed in fixation buffer (1% paraformaldehyde, 2.5% glutaraldehyde, and 0.1 M phosphate buffer pH 7.2) at room temperature for 1 h, and left for fixation overnight at 4°C (*40*). Samples were washed five times in 0.1 M phosphate buffer and then fixed in 1% (wt/vol) osmium tetroxide with 0.1 M phosphate buffer at 4 °C for 1.5 h. The samples were dehydrated in acetone with a graded series of concentrations and embedded in Spurr's resin (SPI supplies). Ultrathin sections were collected on 200 mesh nickel grids coated with Formvar and stained with 1% (w/v) lead citrate and uranyl acetate. Sections were examined with a JEM-1400 transmission electron microscope at an accelerating voltage of 80 kV.

Data analysis

Differences in gene expression levels, *C*Las levels, JA and JA-Ile levels, and relative enrichment fold were determined using Student's *t*-tests for comparing two treatments or two lines. Differences in gene expression levels, relative LUC activity, relative GUS activity, JA and JA-Ile levels, *C*Las levels and relative fluorescence intensity were determined using one-way ANOVA, followed by Duncan's multiple range test for significant differences among different lines or different treatments.

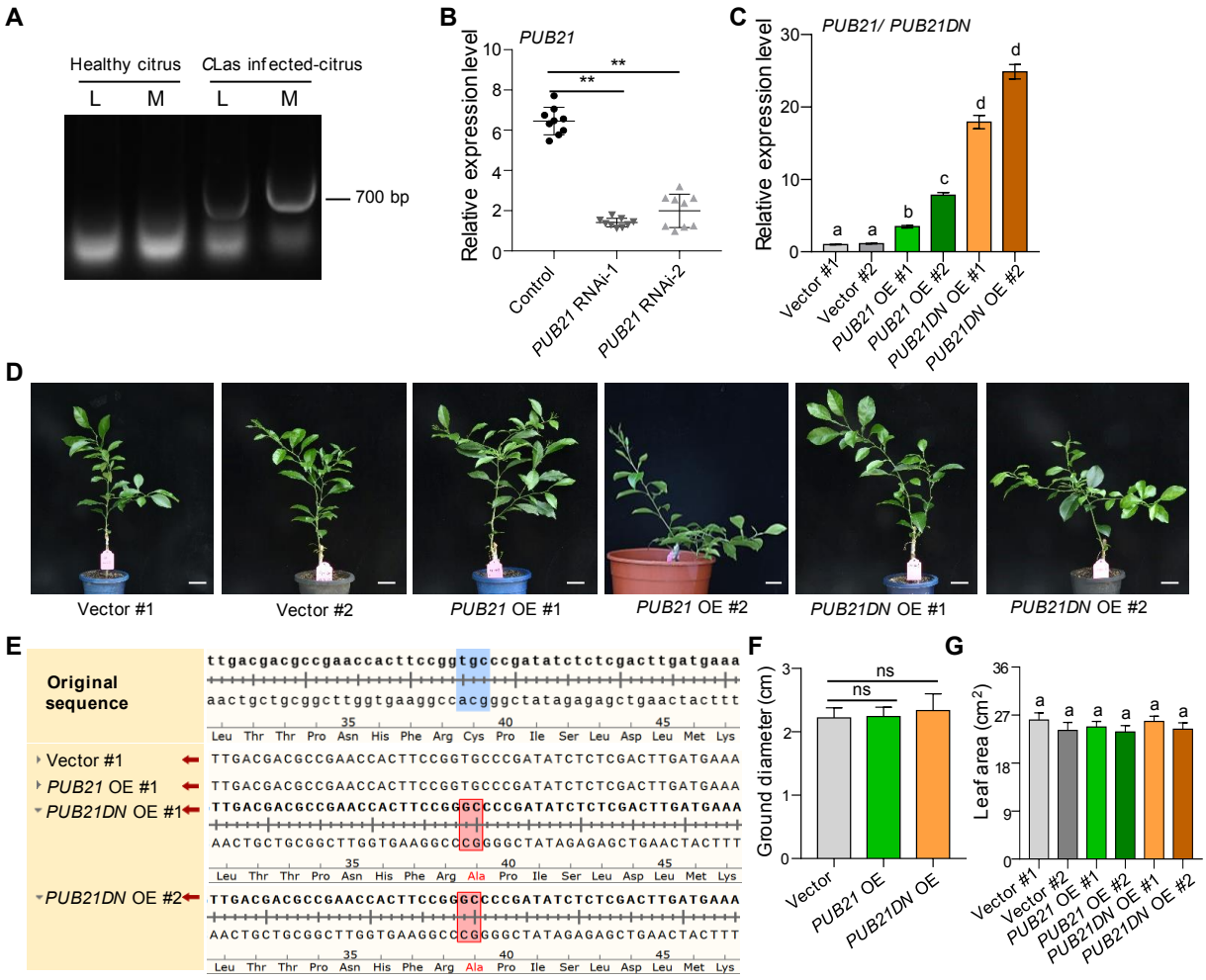


Fig. S1. Stable transgenic citrus trees overexpressing *PUB21* or *PUB21DN*. (**A**) The *C*Las infection in citrus was detected by RT-PCR with A2 and J5 primers. "L" indicates citrus leaves, and "M" indicates citrus midribs. (**B**) Relative *PUB21* expression level in *PUB21*-silencing citrus leaves. Values are means ± SEMs (n=9) (**$P < 0.01$; Student's *t*-test). (**C**) Relative *PUB21* or *PUB21DN* expression levels in transgenic citrus lines. Values are means ± SEMs (n=3). Lowercase letters indicate significant differences among different citrus lines ($P < 0.05$; Duncan's multiple comparisons test). (**D**) The phenotype of transgenic overexpressing *PUB21* and *PUB21DN* in *C. sinensis*. Scale bars, 2.5 cm. (**E**) Sequencing validation of transgenic overexpressing *PUB21* and *PUB21DN* citrus. (**F**) Calculate the ground diameter of two-year old transgenic citrus plants. Values are means ± SEMs (n=3) (ns, no significant differences; Student's *t*-test). (**G**) The leaf area of two-year old transgenic citrus plants. Values are means ± SEMs (n=30). Lowercase letters indicate significant differences among different treatments ($P < 0.05$; Duncan's multiple comparisons test).

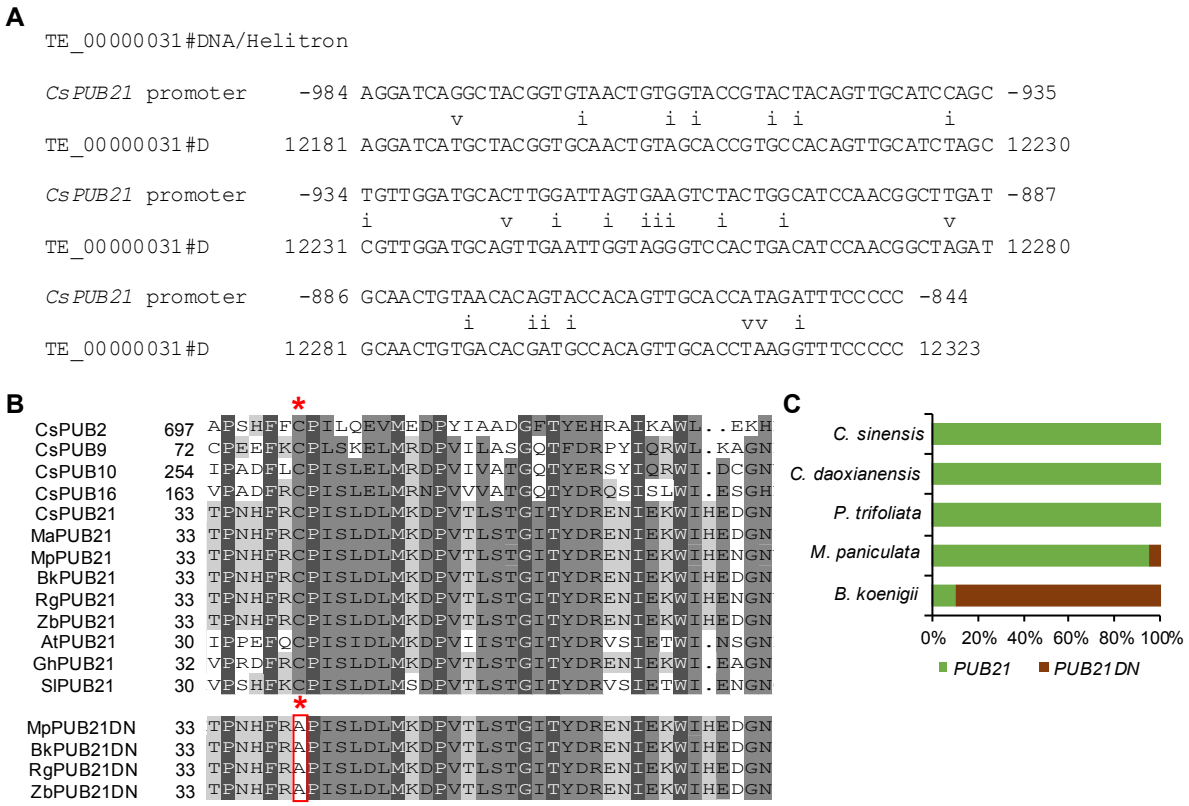


Fig. S2. Two major *PUB21* genomic variations exist in genomes of Citrus and distant relatives. (**A**) A helitron insertion is located in the PUB21 promoter of *C. sinensis*. The genome of *C. sinensis* was annotated with transposons using the EDTA method. The annotated transposon library was aligned with a 150-bp insertion sequence in the *CsPUB21* promoter using the Repeatmaker method. (**B**) Amino acid alignment of a short sequence of the conserved U-box motif in *PUBs* in different species. Numbers refer to the positions of the first amino acid in the sequence. The asterisk indicates a conserved cysteine residue. Cs: *Citrus sinensis*; Mp: *Murraya paniculata*; Bk: *Bergera koenigii*; Ma: *Microcitrus australasica*; Zb: *Zanthoxylum bungeanum*; Rg: *Ruta graveolens*; At: *Arabidopsis thaliana*; Gh: *Gossypium hirsutum*; Sl: *Solanum lycopersicum.* (**C**) The expression ratio of *PUB21* and *PUB21DN* in Rutaceae species.

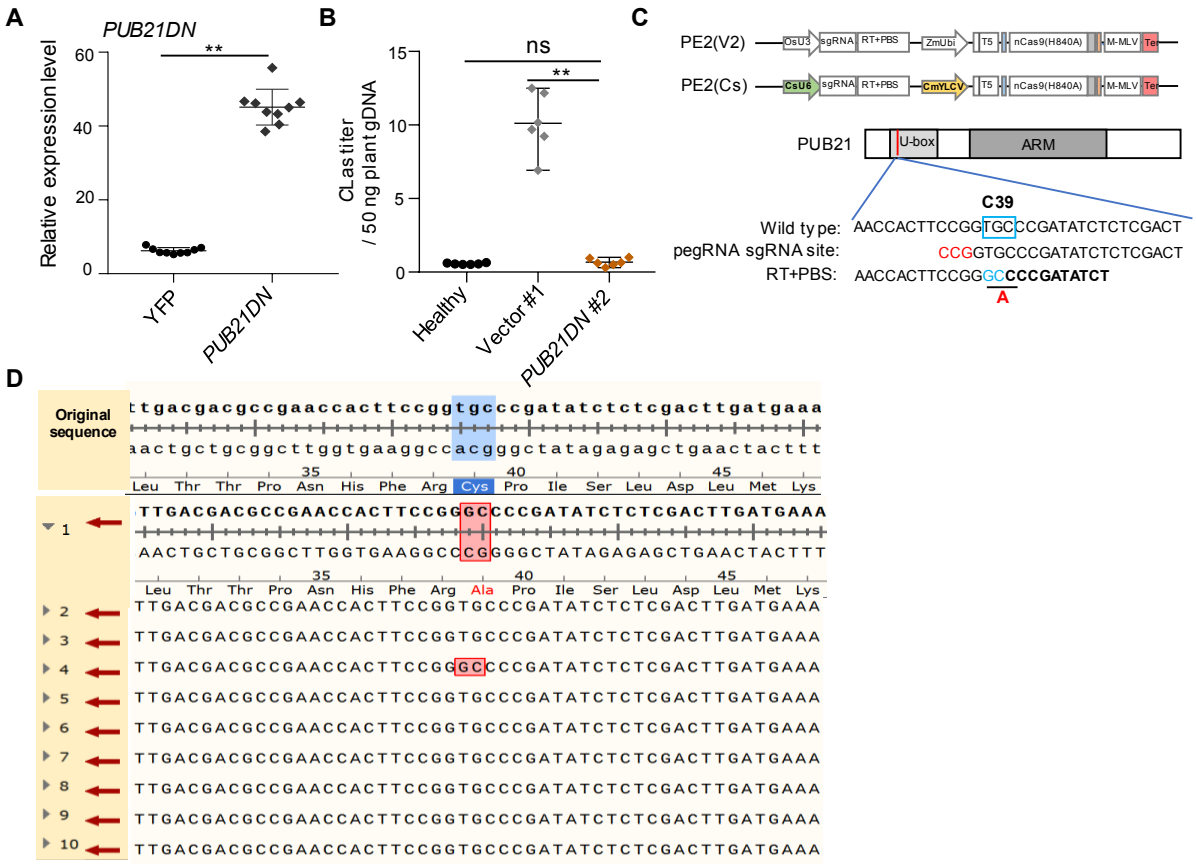


Fig. S3. The ectopic expression of *PUB21DN* enhances citrus resistance to HLB. (**A**) Relative *PUB21DN* expression level in *YFP*-expressing and transient overexpressing *PUB21DN* citrus leaves. Values are means ± SEMs (n=9) (**$P < 0.01$; Student's *t*-test). (**B**) The *C*Las amount in *PUB21DN* transgenic citrus delivered by psyllid vectors. Values are means ± SEMs (n=6) (ns, no significant differences; **$P < 0.01$; Student's *t*-test). (**C**) Schematic diagram of Prime-editing on PUB21 locus. The PAM and the desired insertion are highlighted in red and blue, respectively. The PBS is shown in bold. (**D**) Representative genotypes at the peg-CsPUB21 site as analyzed using sequencing in *C*Las-infected citrus protoplasts. The desired edit is highlighted in red.

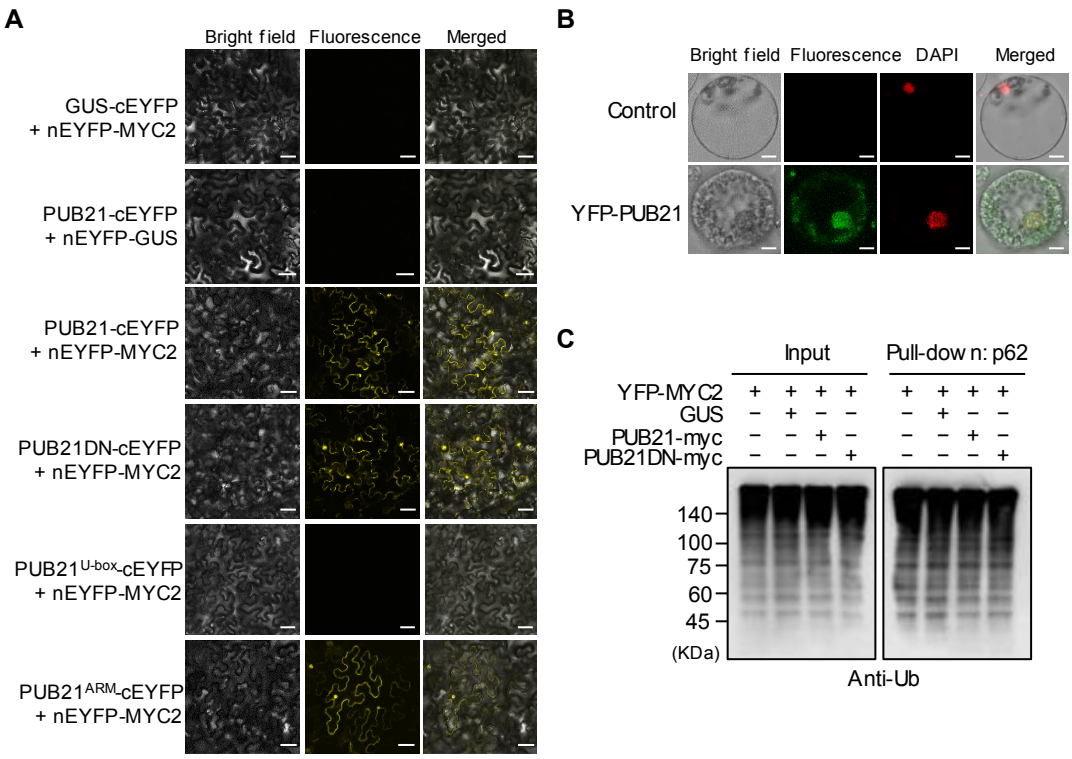


Fig. S4. MYC2 interacts with the ARM domain of PUB21. (**A**) BiFC analysis of MYC2 interaction with PUB21 deletion derivatives. Scale bars, 50 μm. (**B**) Subcellular localization of PUB21 in *C. sinensis* protoplasts. The 4′,6-diamidino-2-phenylindole (DAPI) staining is used to confirm the nuclei of cells. Scale bars, 5 μm. (**C**) Affinity purification of polyubiquitinated YFP-MYC2 was carried out from total protein extracts. Anti-Ub was used to detect total ubiquitinated-proteins.

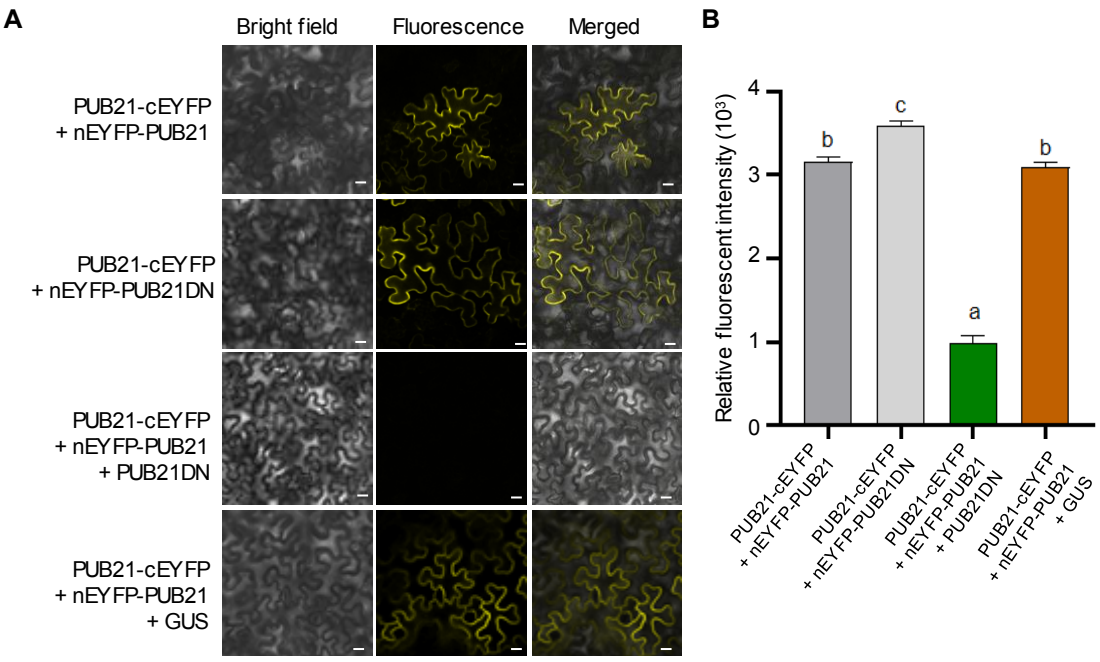


Fig. S5. PUB21DN competes for the interaction with PUB21. (**A**) Modified BiFC competition assays. The EYFP fluorescences were detected using co-expression of PUB21-cEYFP + nEYFP-PUB21 with GUS or PUB21DN. Scale bars, 50 μm. (**B**) Quantification of the effects of PUB21DN on PUB21 dimerization. Values are means ± SEMs (n=15). Lowercase letters indicate significant differences among different treatments ($P < 0.05$; Duncan's multiple comparisons test).

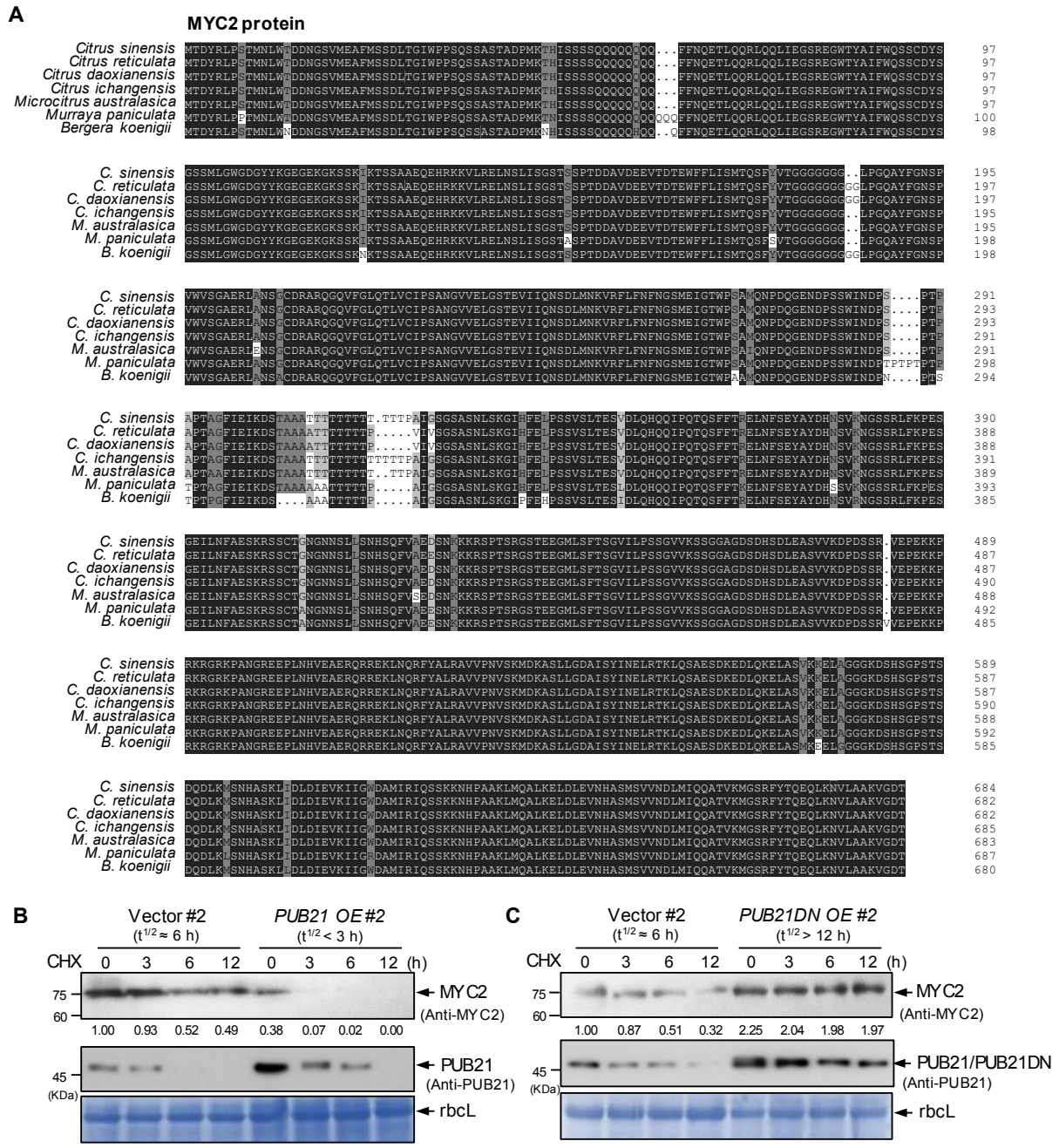


Fig. S6. PUB21 promotes MYC2 degradation. (**A**) The alignment of the amino acid sequences of MYC2 from Rutaceae species. (**B** and **C**) Degradation rate of MYC2 in transgenic citrus overexpressed *PUB21* and *PUB21DN*. Transgenic overexpressing *PUB21* (line 2) (B) and *PUB21DN* (line 2) citrus leaves (C) were incubated with 200 μM CHX. Proteins were extracted and detected for the indicated time. Band intensities were measured using ImageJ software and normalized to that of loading rbcL protein. The normalized value at 0 h in vector was set to 1.00.

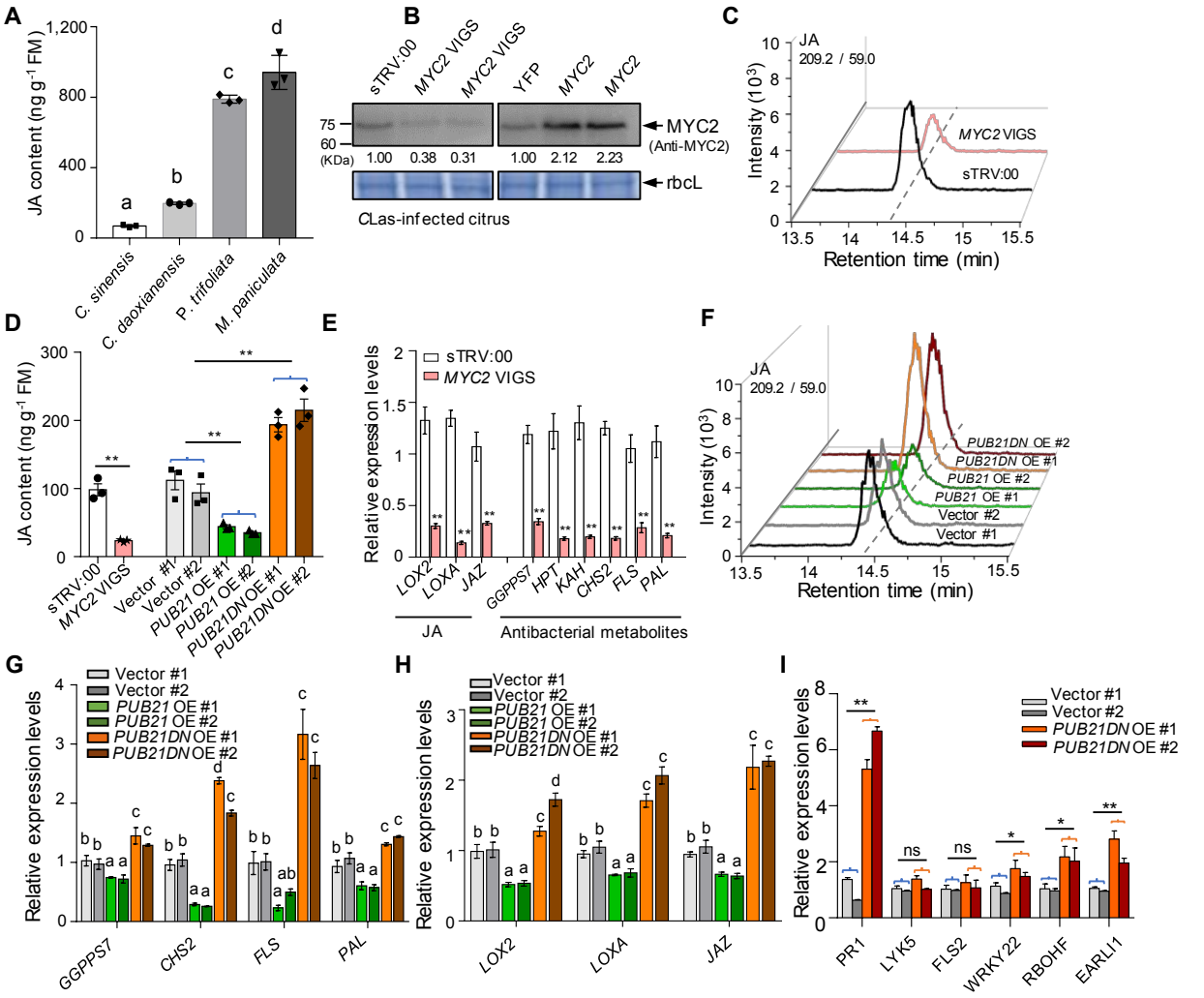


Fig. S7. PUB21DN promotes JA-mediated secondary metabolism and immunity. (**A**) The levels of endogenous JA in different Rutaceae species. (**B**) Detection of MYC2 levels in *MYC2*-silencing and transient overexpressing *MYC2* citrus. *C*Las-infection citrus was used here. (**C**) HPLC–MS analysis of endogenous JA in *MYC2*-silencing citrus. (**D**) Quantification of endogenous JA in *MYC2*-silencing citrus and transgenic-citrus lines. (**E**) Relative expression levels of JA-mediated defense genes in *MYC2*-silencing citrus. JA-related biosynthetic and regulatory genes include *LOX2*, *LOXA* and *JAZ*. Antibacterial metabolite biosynthesis-related genes include *geranylgeranyl pyrophosphate synthase 7* (*GGPPS7*), *homogentisate phytyltransferase* gene (*HPT*) and *ent-kaurenoic acid hydroxylase* (*KAH*) involved in isoprenoid synthesis, *chalcone synthase 2* (*CHS2*) and *flavonol synthase* (*FLS*) involved in flavonoid synthesis, and *phenylalanine ammonia-lyase* (*PAL*) involved in phenylpropanoid metabolism. (**F**) HPLC–MS analysis of endogenous JA in transgenic citrus lines. (**G**) Relative expression levels of antibacterial metabolite-related genes in overexpressing *PUB21* and *PUB21DN* transgenic lines. (**H**) Relative expression levels of JA-related genes in the transgenic citrus lines. In figures A, G and H, values are means ± SEMs (n=3). Lowercase letters indicate significant differences among different citrus lines ($P < 0.05$; Duncan's multiple comparisons test). (**I**) Relative expression levels of resistance-related genes in the *PUB21DN* OE transgenic citrus lines. In figures D, E and I, values are means ± SEMs (n=3) (ns, no significant differences; *$P < 0.05$; **$P < 0.01$; Student's *t*-test).

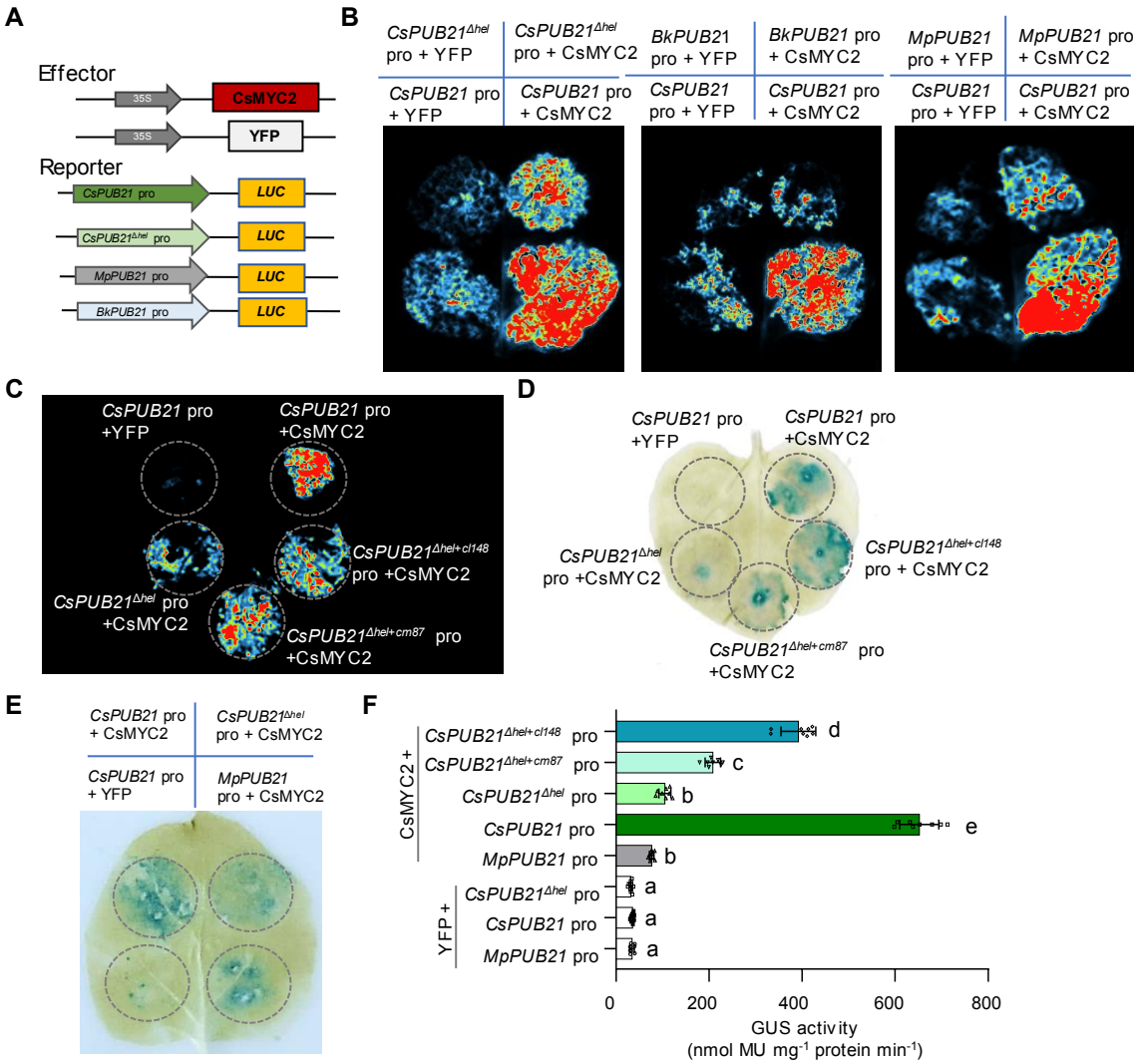


Fig. S8. MYC2 positively regulates the *PUB21* expression. (**A**) Schematic diagram showed effector and reporter constructs in Figure 3B. *CsPUB21* promoter: *luciferase* (*LUC*), *CsPUB21*$^{\Delta hel}$ promoter: *LUC*, *MpPUB21* promoter: *LUC* and *BkPUB21* promoter: *LUC* were used as reporter constructs. *35S* promoter-driven *YFP* and *MYC2* were used as effector constructs. (**B** and **C**) Luciferase imaging of MYC2 transcriptional activity on different *PUB21* promoters in *N. benthamiana*. (**D** and **E**) The different *PUB21* promoter activity in a *YFP* and *MYC2* expressing leaf by β-glucuronidase (GUS) staining. (**F**) The transcriptional activity of MYC2 on the *PUB21* promoters. Values are means ± SEMs (n=8). Lowercase letters indicate significant differences among different columns ($P < 0.05$; Duncan's multiple comparisons test).

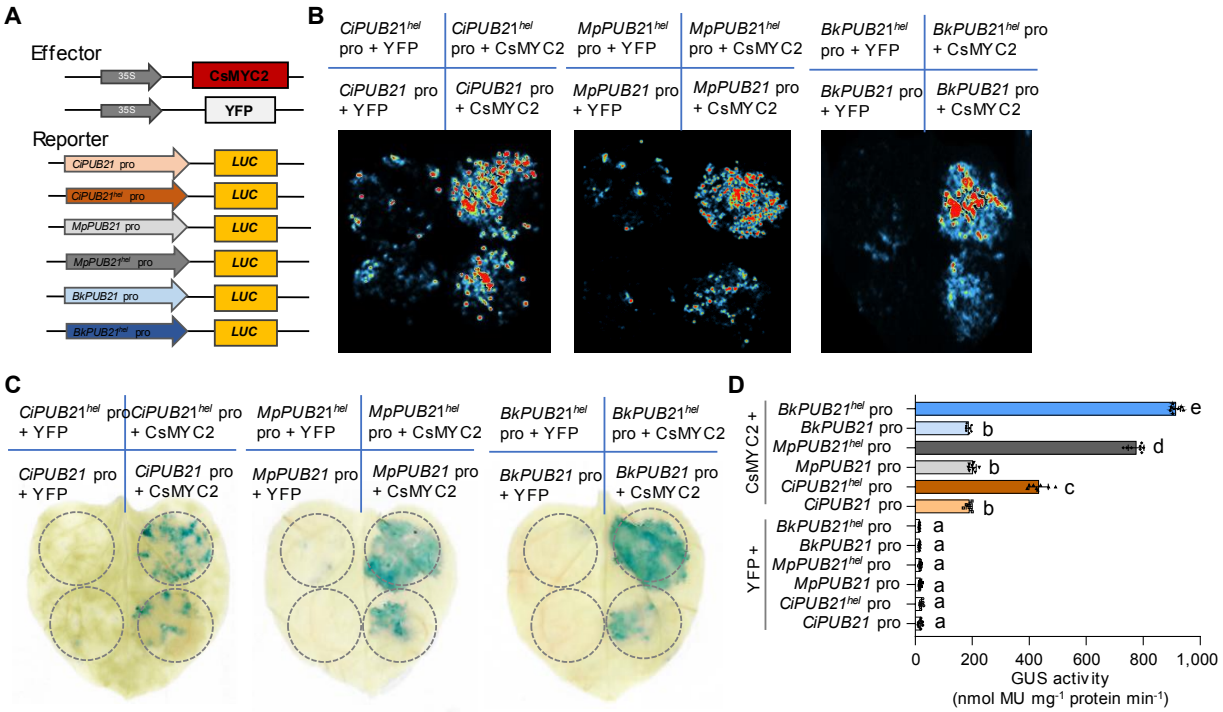


Fig. S9. Helitron insertion in the *PUB21* promoter enhances the *PUB21* expression. (**A**) Schematic diagram shows the effector and reporter constructs used in Figure 3C. Different promoters driven-*LUC* were used as reporter constructs. *35S* promoter-driven *YFP* and *MYC2* were used as effector constructs. (**B**) Luciferase imaging of MYC2 transcriptional activity on different *PUB21* promoters in *N. benthamiana*. (**C**) The different *PUB21* promoter activity in a *YFP* and *MYC2* expressing leaf by GUS staining. (**D**) The transcriptional activity of MYC2 on the *PUB21* promoters. Values are means ± SEMs (n=8). Lowercase letters indicate significant differences among different columns ($P < 0.05$; Duncan's multiple comparisons test).

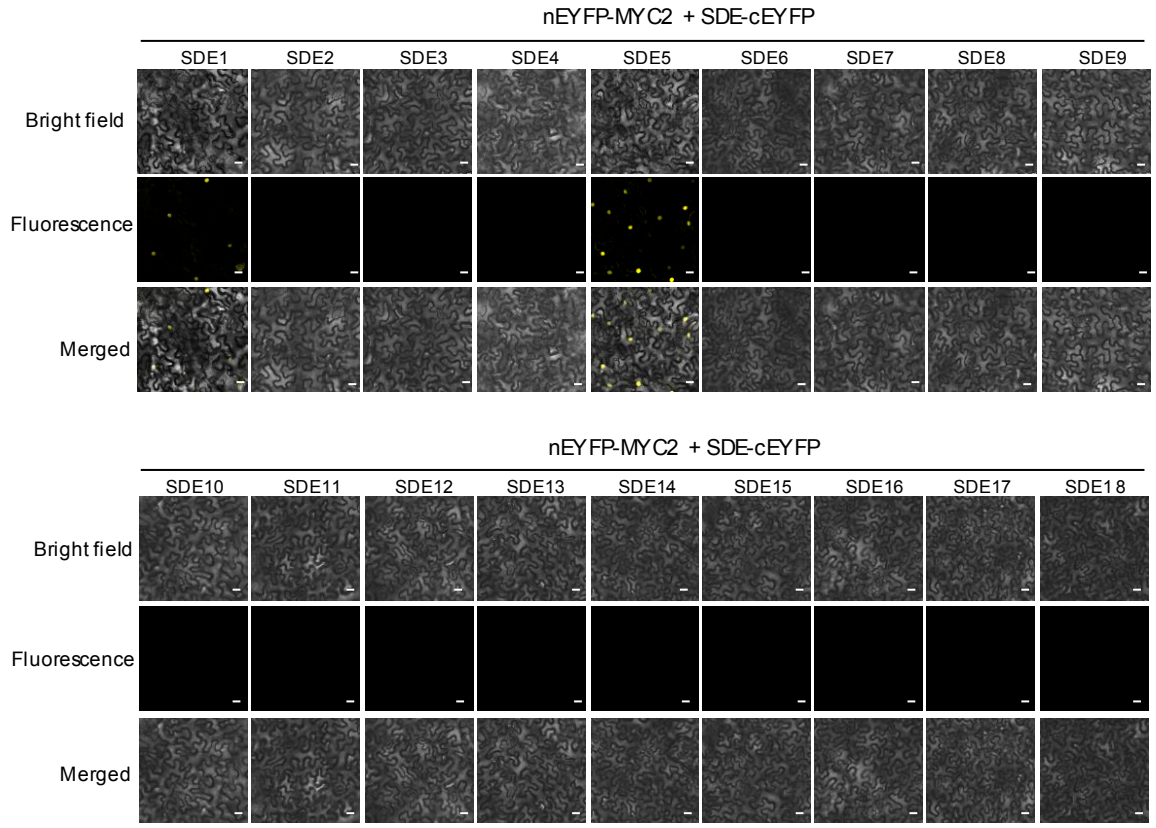


Fig. S10. SDE1 and SDE5 interacts with MYC2. BiFC analysis of SDEs interaction with MYC2 in planta. SDE-cEYFP and nEYFP-MYC2 were transiently expressed in *N. benthamiana* leaf epidermal cells via agroinfiltration. Scale bars, 50 μm.

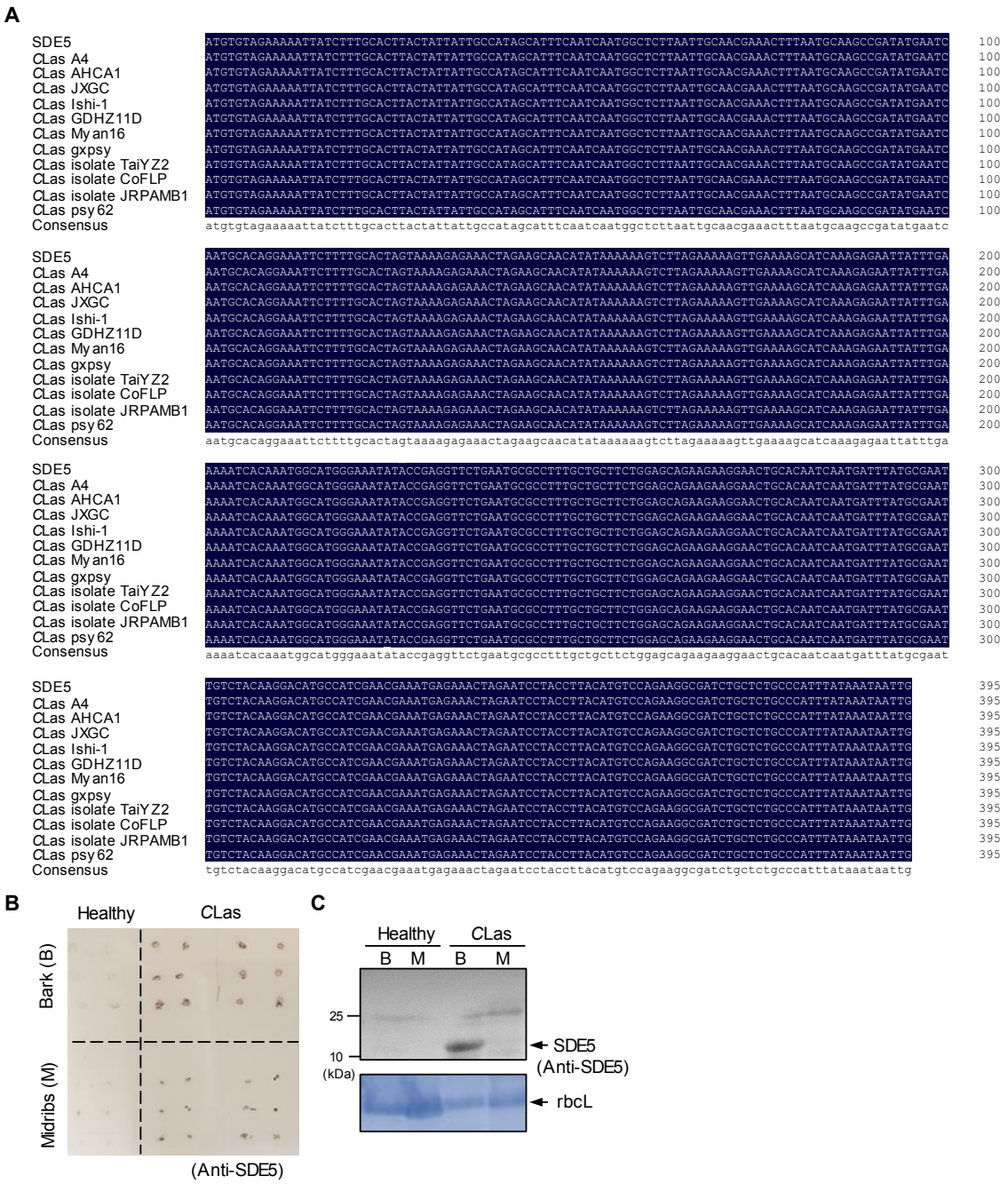


Fig. S11. SDE5 is secreted into phloem cells in *C*Las-infected citrus. (**A**) SDE5 sequences from eleven *C*Las strains are 100% identical. The alignment of the nucleotide sequences of SDE5 from various *C*Las strains, including A4, gxpsy, JXGC, and GDHZ11D from China, AHCA1 from California, JRPAMB1 and Psy62 from Florida, Ishi-1 from Japan, Myan16 form Myanmar, CoFLP from Colombia, TaiYZ2 from Thailand. (**B**) Protein abundance of SDE5 was determined in the bark (B) and midribs (M) tissues in healthy citrus and *C*Las-infected citrus. Freshly cut stems were printed onto nitrocellulose membranes, which were then incubated with the anti-SDE5 antibody. (**C**) Western blotting confirming SDE5 protein accumulation in the phloem sap of healthy citrus and *C*Las-infected citrus branches. SDE5 protein was detected by western blot using an SDE5-specific antibody.

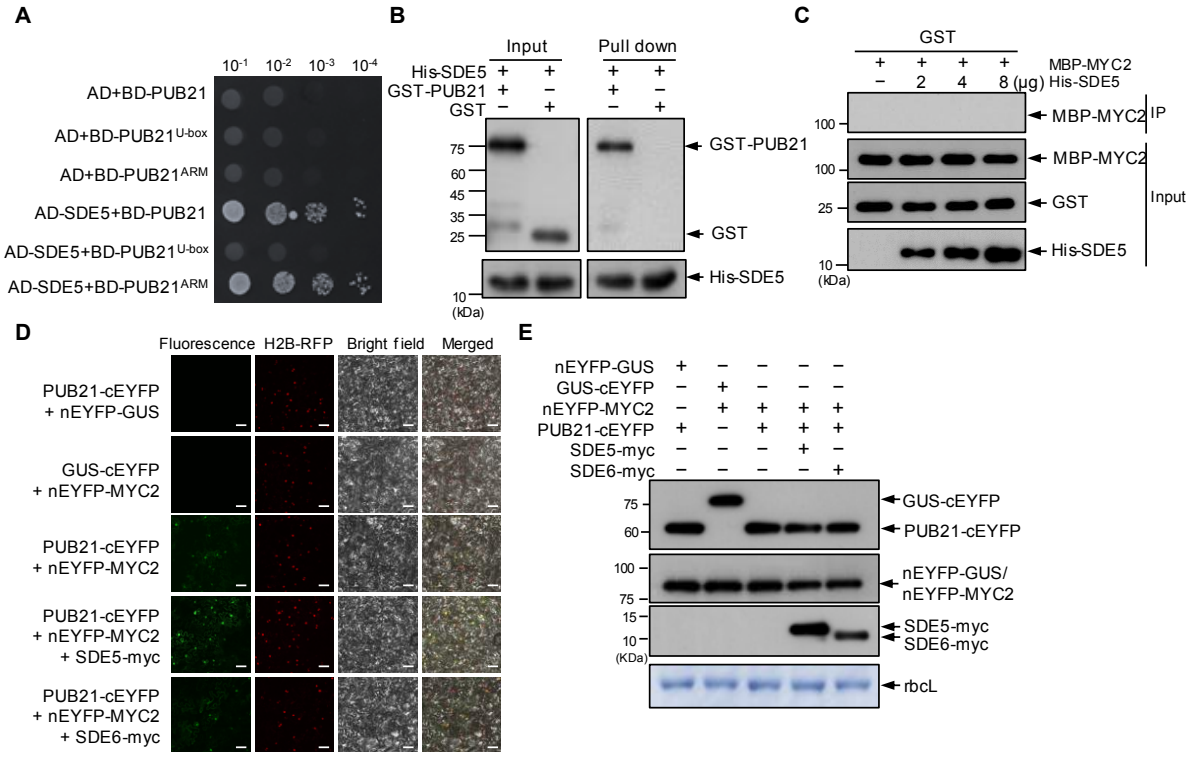


Fig. S12. SDE5 promotes the PUB21-MYC2 interaction. (**A**) Interaction between SDE5 and PUB21 in the yeast two-hybrid system. Yeast strain AH109 cotransformed with the indicated plasmids was spotted on SD-Leu-Trp-His medium. (**B**) Pull-down assays between SDE5 and PUB21. (**C**) Protein competition pull-down assay. The indicated protein amount of His-SDE5 was mixed with 2 μg of MBP-MYC2 and pulled down by 2 μg of GST. The associated proteins were detected by immunoblots using anti-MBP antibodies. (**D**) Modified BiFC competition assays. The EYFP fluorescences were detected using co-expression of PUB21-cEYFP + nEYFP-MYC2 with SDE5 or SDE6. Scale bars, 50 μm. (**E**) Detection of protein levels in modified BiFC assay.

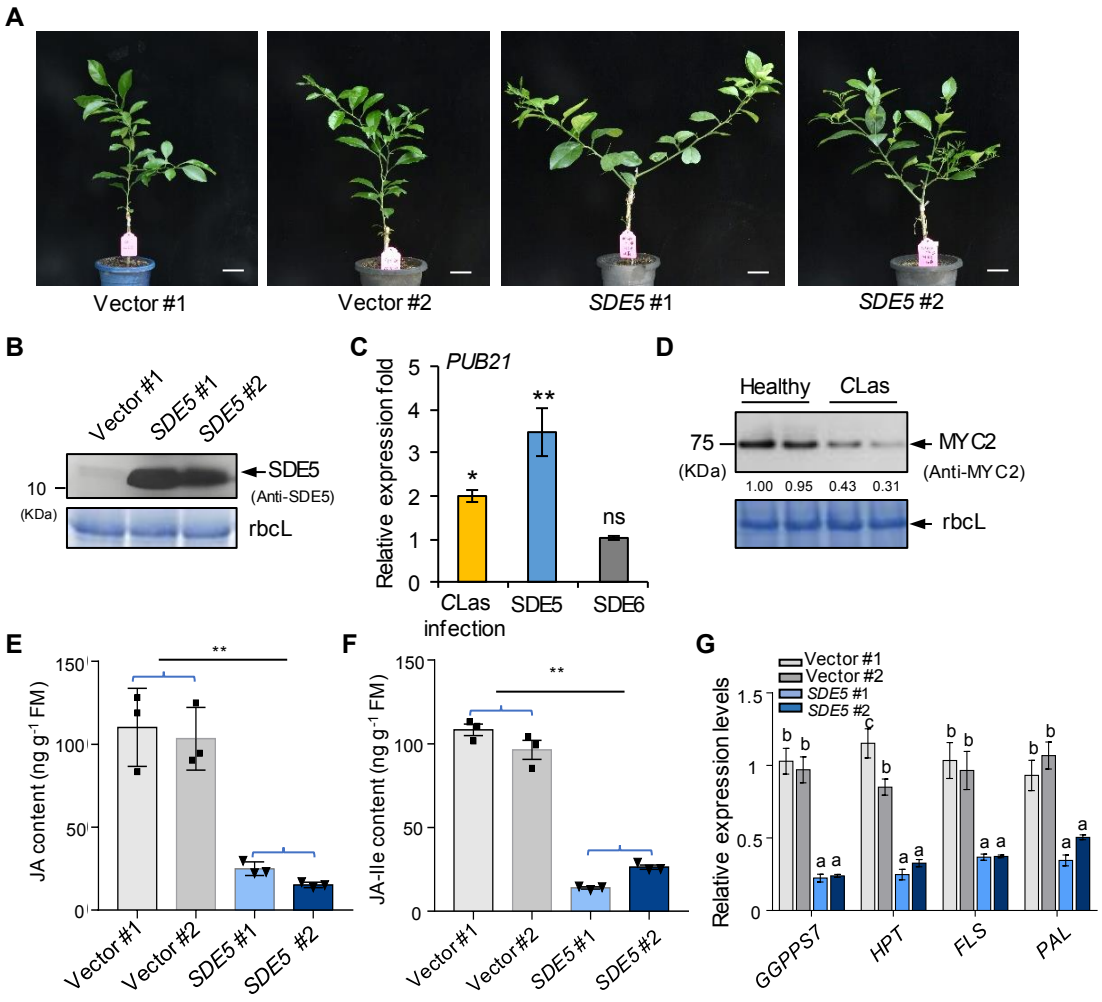


Fig. S13. SDE5 negatively regulates JA response. (**A**) Phenotype of *SDE5*-transgenic citrus trees. Scale bars, 2.5 cm. (**B**) Detection of SDE5 protein levels in *SDE5*-transgenic citrus trees. (**C**) Relative *PUB21* expression levels upon *C*Las infection or SDEs expression. Values are means ± SEMs (n=3) (ns, no significant differences; **P* < 0.05; ***P* < 0.01; Student's *t*-test). (**D**) MYC2 protein accumulation in the healthy and *C*Las-infected citrus. (**E** and **F**) Quantification of endogenous JA (E) and JA-Ile (F) in vector control and two *SDE5*-transgenic citrus lines. Values are means ± SEM (n=3) (***P* < 0.01; Student's *t*-test). (**G**) Relative expression levels of antibacterial metabolism-related genes in *SDE5*-transgenic citrus lines. Values are means ± SEMs (n=3). Lowercase letters indicate significant differences among different citrus lines (*P* < 0.05; Duncan's multiple comparisons test).

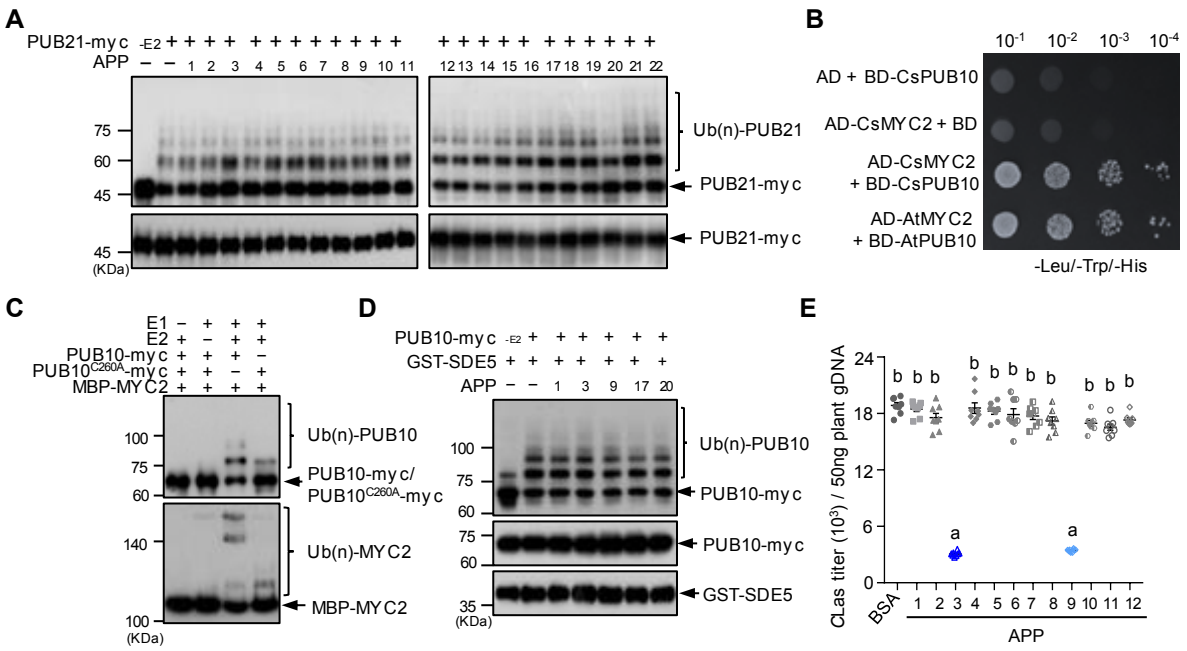


Fig. S14. APP3 does not affect the E3 ligase activity of PUB10. (**A**) Effects of peptides on the auto-ubiquitination of PUB21 protein. These peptides can bind to PUB21 protein by peptide-protein molecular docking. (**B**) The interaction of citrus PUB10 with MYC2 in the yeast two-hybrid system. (**C**) MYC2 is the substrate of PUB10 E3 ligase. Polyubiquitination of MBP-MYC2 was assayed in the presence of AtUBA1 (E1), AtUBC8 (E2), PUB10 (E3), and ubiquitin. $PUB10^{C260A}$ mutant protein was used as a negative control. (**D**) Effects of APP peptides on the auto-ubiquitination of PUB10 protein with SDE5. (**E**) The *C*Las amount in *C*Las-infected citrus leaves treated by APPs for 48h. APPs were injected into *C*Las-infected citrus leaves using a syringe. BSA was used as the negative control. Values are means ± SEMs (n=8). Lowercase letters indicate significant differences among different citrus lines ($P < 0.05$; Duncan's multiple comparisons test).

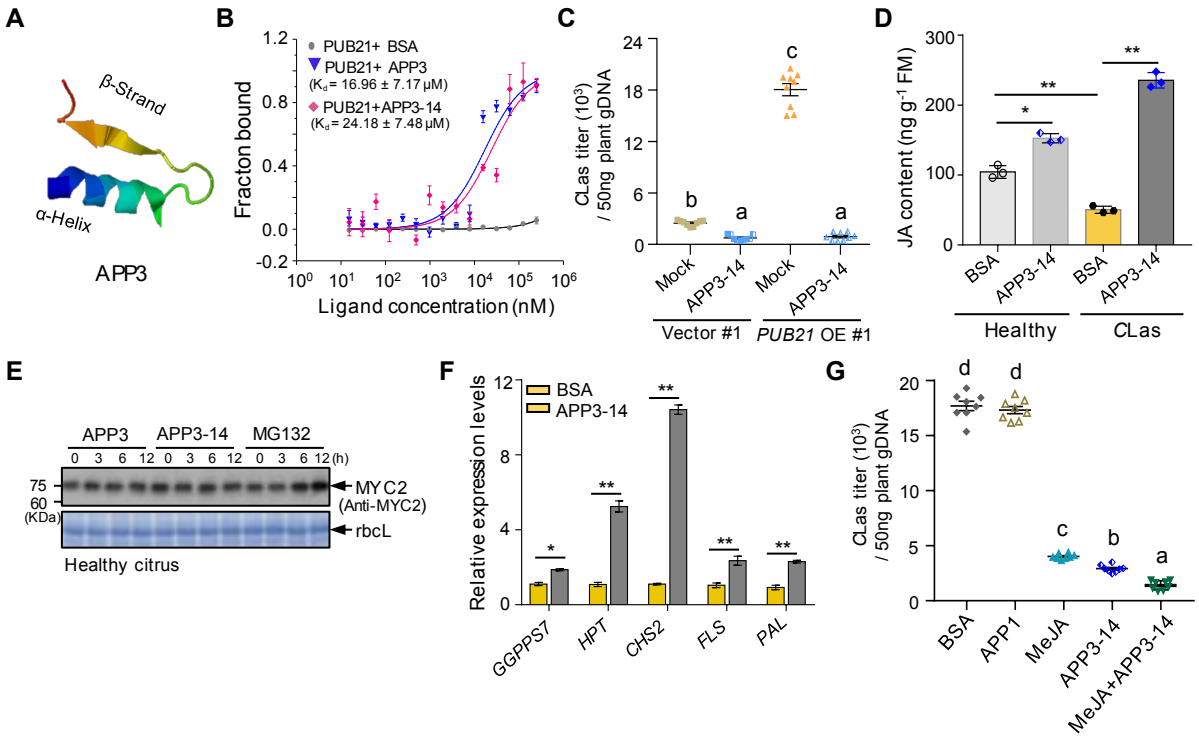


Fig. S15. APP3-14 promotes JA-mediated defense. (**A**) Model structure of APP3 containing one α-Helix and one β-Strand. (**B**) APP3 and APP3-14 directly bind to PUB21 through microscale thermophoresis (MST) assay. BSA was used as the negative control. The Kd was quantified to estimate the binding affinity of direct interactions. Values are means ± SEMs (n=3). (**C**) The *C*Las amount in citrus leaves with or without treated by 1 μM APP3-14 peptides to *PUB21* overexpressing lines. Values are means ± SEMs (n=9). Lowercase letters indicate significant differences among different treatments ($P < 0.05$; Duncan's multiple comparisons test). (**D**) Quantification of endogenous JA in APP3-14 treated healthy and *C*Las-infected citrus leaves. Values are means ± SEMs (n=3) (*$P < 0.05$; **$P < 0.01$; Student's *t*-tests). (**E**) Effects of APP3 and APP3-14 peptides on MYC2 accumulation in healthy citrus. (**F**) Relative expression levels of antibacterial metabolite-related genes in *C*Las-infected citrus upon APP3-14 treatment. Values are means ± SEM (n=3) (*$P < 0.05$; **$P < 0.01$; Student's *t*-tests). (**G**) The *C*Las amount in *C*Las-infected *C. sinensis* leaves treated by MeJA and APP3-14 for 48 h. 10 μM MeJA and 1 μM APP3-14 were injected into *C*Las-infected citrus leaves using a syringe. Values are means ± SEMs (n=8). Lowercase letters indicate significant differences among different treatments ($P < 0.05$; Duncan's multiple comparisons test).

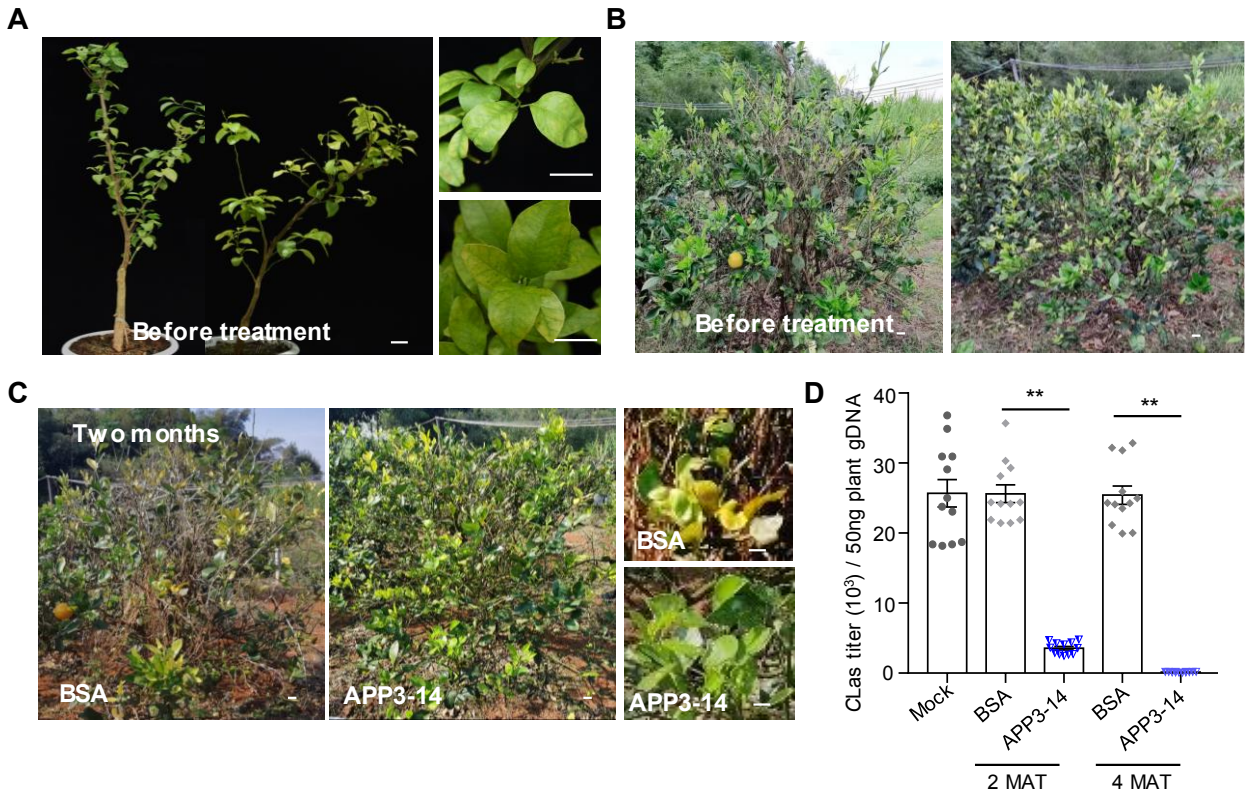


Fig. S16. APP3-14 suppresses HLB symptoms. (**A**) The image of *C*Las-infected *C. sinensis* before treatment in the greenhouse. Scale bars, 2 cm. (**B**) The image of *C*Las-infected *C. sinensis* trees before treatment in field traits. Scale bars, 3 cm. (**C**) Symptoms of symptomatic leaves/total *C. sinensis* trees were investigated after treatment with 1 μM BSA or 1 μM APP3-14 peptide for two months in field traits. (**D**) The *C*Las amount in HLB-positive trees treated by APP3-14 or BSA in field trials. The *C*Las titers were evaluated by 3 branches in each tree with four treated-trees. MAT, month after treatment. Values are means ± SEMs (n=12) (**$P < 0.01$; Student's *t*-tests).

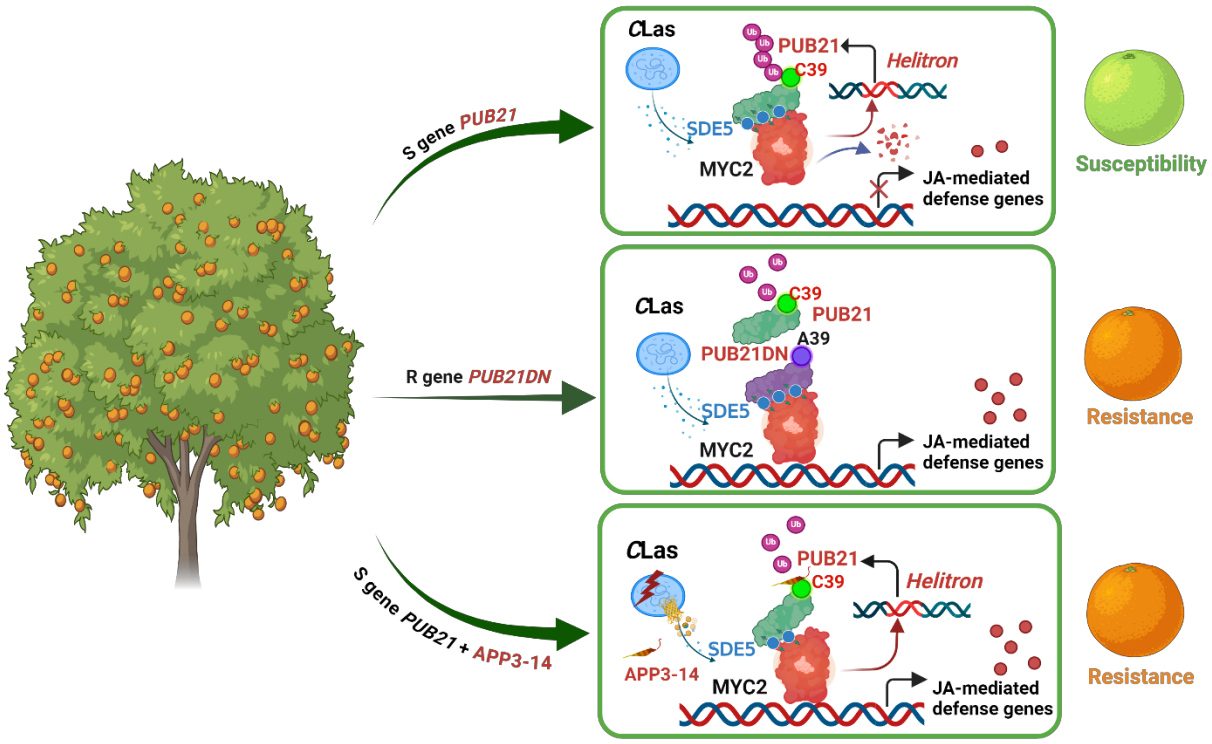


Fig. S17. The working model of targeted MYC2 stabilization conferring citrus Huanglongbing resistance. In HLB-sensitive species, a helitron insertion in the *PUB21* promoter recruits more MYC2 for enhancing *PUB21* expression. The E3 ligase PUB21 ubiquitinates and degrades MYC2 protein to suppress JA-mediated responses, thereby acting as an *S* gene to promote HLB in citrus. In HLB-resistant species, MYC2 less feedback activates *PUB21* expression due to no helitron insertion in the *PUB21* promoter. PUB21 and a dominant-negative paralog PUB21^{C39A} (PUB21DN) genomic coupling exists in natural HLB-resistance species. PUB21DN impairs E3 activity both itself and PUB21 and thereby promotes MYC2 stability, conferring resistance against HLB infection. In *C*Las-infection plants, SDE5 suppresses JA-mediated defense by enhancing PUB21-mediated degradation of MYC2. Thus, SDE5 promotes citrus susceptibility to HLB. Importantly, when the application of APP3-14 as an anti-proteolysis peptide-based inhibitor treats *C*Las-infection citrus, this peptide together with SDE5 inhibits PUB21 activity by intensifying to bind C39 of PUB21. APP3-14 has a dual function that promotes JA-mediated responses and directly breaks *C*Las cells, thereby preventing and controlling HLB.

Table S1. Plant U-box (*PUB*) genes in *Citrus sinensis*

Name	Locus tag
PUB1	Cs_ont_4g008460
PUB2	Cs_ont_4g011280
PUB3	Cs_ont_2g034110
PUB4	Cs_ont_1g024110
PUB5	Cs_ont_3g000850
PUB6	Cs_ont_7g022460
PUB7	Cs_ont_1g026830
PUB8	Cs_ont_8g002740
PUB9	Cs_ont_7g015800
PUB10	Cs_ont_2g012600
PUB11	Cs_ont_8g000560
PUB12	Cs_ont_5g004900
PUB13	Cs_ont_8g002840
PUB14	Cs_ont_7g006740
PUB15	Cs_ont_8g028960
PUB16	Cs_ont_6g013270
PUB17	Cs_ont_1g022260
PUB18	Cs_ont_7g024750
PUB19	Cs_ont_5g046090
PUB20	Cs_ont_3g001030
PUB21	Cs_ont_2g014020
PUB22	Cs_ont_6g009510
PUB23	Cs_ont_8g009320
PUB24	Cs_ont_6g009500
PUB25	Cs_ont_1g005320
PUB26	Cs_ont_8g024850
PUB27	Cs_ont_1g004560
PUB28	Cs_ont_9g006760
PUB29	Cs_ont_3g007590
PUB30	Cs_ont_7g001820
PUB31	Cs_ont_5g021070
PUB32	Cs_ont_9g025970
PUB33	Cs_ont_5g003240
PUB34	Cs_ont_6g000550
PUB35	Cs_ont_9g022930
PUB36	Cs_ont_7g015790
PUB37	Cs_ont_8g009310
PUB38	Cs_ont_5g046370
PUB39	Cs_ont_5g015760
PUB40	Cs_ont_6g002280
PUB41	Cs_ont_1g023420
PUB42	Cs_ont_8g013920

Table S2. Putative proteins interact with PUB21

Locus tag	Interaction protein annotation	Gene
XP_006491734.1	Tanscription factor MYC2-like	MYC2
XP_006492535.1	Heat stress transcription factor A-5	HSF A-5
XP_006469048.1	E3 ubiquitin-protein ligase ATL41-like	ATL41L
XP_006491633.1	Indole-3-acetic acid-amido synthetase GH3.10	GH3.10
XP_006484609.1	WEB family protein	WEB1
XP_006487153.1	Guanine nucleotide-binding protein subunit beta-2	RACK2
XP_006490465.1	Sorbitol dehydrogenase-like	SDL1

Table S3. *Candidatus* Liberibacter asiaticus (*C*Las) putative sec-delivered effectors (SDEs) with smaller than 20 kDa

Gene name	Locus tag	Length (bp)
SDE1	CLIBASIA_05315	465
SDE2	CLIBASIA_04735	489
SDE3	CLIBASIA_00420	483
SDE4	CLIBASIA_04410	369
SDE5	CLIBASIA_02470	396
SDE6	CLIBASIA_04580	351
SDE7	CLIBASIA_01330	393
SDE8	CLIBASIA_00755	375
SDE9	CLIBASIA_00235	417
SDE10	CLIBASIA_00185	207
SDE11	CLIBASIA_00530	294
SDE12	CLIBASIA_00525	294
SDE13	CLIBASIA_04055	207
SDE14	CLIBASIA_04030	267
SDE15	CLIBASIA_04025	291
SDE16	CLIBASIA_05320	258
SDE17	CLIBASIA_04530	258
SDE18	CLIBASIA_04040	480

Table S4. Putative AMPs come from human gut microbiome

AMPdiscover		
sORFs	AMPs	rep-AMPs
122826	40837	14950

Table S5. Randomly select synthetic peptides for inhibiting PUB21 E3-ligase activity.

Peptides	Molecular
APP1	LBBUAG_103173
APP 2	VNLJVH_73940
APP 3	QYMFSE_35673
APP 4	NCOYWW_169977
APP 5	GFCDWA_231836
APP 6	NJYHFZ_170965
APP 7	XPFSLP_86513
APP 8	XVOCUO_150422
APP 9	HGCRCD_40783
APP10	CZCLBL_46867
APP11	EPGFKU_60425
APP12	CDWDUZ_89601
APP13	CPUYZI_150467
APP14	RCDHJQ_47762
APP15	OKSFPA_157598
APP16	UTBIDH_87961
APP17	JCGWKW_183486
APP18	HUPCUU_22270
APP19	LRQQPD_85619
APP20	FZLXJR_36596
APP21	XBTSPW_106947
APP22	ELVHXQ_110688

Table S6. Primers were used in this study.

CDS cloning

Primers names	Primer Sequence (5’-3’)
PUB21-Fw(BamHI)	CAAGGGATCCATGATTTTGTCATGGAAAAGAC
PUB21-Rv(StuI)	CAAGAGGCCTAAACGGCCTTTTCAGGTCCT
MYC2-Fw(BamHI)	CAAGGGATCCATGACGGACTACCGGTTACC
MYC2-Rv(StuI)	CAAGAGGCCTTTGGGTATCTCCAACTTTGG
SDE5-Fw(BamHI)	CACCAAGCCAGGGATCCATGCTTAATTGCAACGA
SDE5-Rv(StuI)	CAAGAGGCCTTCAATTATTTATAAATGGGC
PUB10-Fw(BamHI)	CAAGGGATCCATGGTCGGCGGTTTCACCCT
PUB10-Rv(StuI)	CAAGAGGCCTCACCTGTTGAAGCTGCGGCA
PUB21-Ubox-Fw(KpnI)	CAAGGGTACCATGATTTTGTCATGGAAAAG
PUB21-Ubox-Rv(XhoI)	CAAGCTCGAGGCCGTCGTGATATTACAAT
PUB21-ARM-Fw(KpnI)	CAAGGGTACCATGTGCAAAAGTGAAGATCAAG
PUB21-ARM-Rv(XhoI)	CAAGCTCGAGAACGGCCTTTTCAGGTCCTT

Mutant cloning

Primers names	Primer Sequence (5’-3’)
PUB21DN-Fw	CCGAACCACTTCCGGGCCCCGATATCTCTCGAC
PUB21DN-Rv	TCTTTCATCAAGTCGAGAGATATCGGGGCC

Gene silence cloning

Primers names	Primer Sequence (5’-3’)
PUB21-RNAi-Fw(KpnI)	CAAGGGTACCGTAGAAAGCGGGGGAGAGATGGAG
PUB21-800RNAi-Rv(XhoI)	CAAGCTCGAGTTAAATAGCGGTTCAATGGC
PUB21-606RNAi-Rv(XhoI)	CAAGCTCGAGCAGCAACGTCAAAGTCGACA
MYC2-VIGS-Fw(BamHI)	CAAGGGATCCCTTCAGCAATTGATTGAAGG
MYC2-VIGS-Rv(XhoI)	CAAGCTCGAGTCAGTGGACCCCAATTCAAC

Promoter cloning

Primers names	Primer Sequence (5’-3’)
CsPUB21 promoter Fw(BamHI)	CCAATTCAGTCGACTGGATCCTCACGCTCAAAGCACTACGA
CsPUB21 promoter Rv(XhoI)	GCTGGGTCTAGATATCTCGAGGCAAGCCAGCAGTATGCAAG
MpPUB21 promoter Fw(BamHI)	CCAATTCAGTCGACTGGATCCGTGGTCTAATATGAACGCCTC
MpPUB21 promoter Rv(XhoI)	GCTGGGTCTAGATATCTCGAGGCAAGCCAGCAATACGCAAG
BkPUB21 promoter Fw(BamHI)	CCAATTCAGTCGACTGGATCCTCGAATGCCTGGCCGATTTT
BkPUB21 promoter Rv(XhoI)	GCTGGGTCTAGATATCTCGAGCCTTGAACAAAAGAAGCAGGCC
CrPUB21 promoter Fw(KpnI)	CAAGGGTACCGTCAAGATTAGCATACACAC
CrPUB21 promoter Rv(XhoI)	CAAGCTCGAGTGTTATATATTGCGCACAAC

Prime editing

Primers names	Primer Sequence (5’-3’)
PUB21-RT-PBS-Fw	TTGTGCAGATGATCCGTGGCGAGTCGAGAGATATCGGGCACGTTTTAG AGCTAGAAATA
PUB21-RT-PBS-Rv	ACGCTGCACTGCAGGCATGCAAGCTTAAAAAAAAGATATCGGGGCCCG GAAGTGGTTGCACCGACTCGGTGCCAC

RT-qPCR

Primers names	Primer Sequence (5’-3’)
A2	TATAAAGGTTGACCTTTCGAGTTT
J5	ACAAAAGCAGAAATAGCACGAACAA
HLBas	GTCGAGCGCGTATGCAATACG
HLBr	GCGTTATCCCGTAGAAAAAGGTAG
COX-qF	GGTATGCCACGTCGCATTCCAGA
COX-qR	GCCAAAACTGCTAAGGGCATTC
SDE5-qF	ATGCTTAATTGCAACGAAAC
SDE5-qR	GTCCTTGTAGACAATTCGCA
PUB21-qF	CGTTGGTCGTCGTCTATCGT
PUB21-qR	AATGGAGACTGCGAACTCCG
MYC2-1 qF	GCGAGCTGTTGTGCCTAATG
MYC2-1 qR	GCCAACTCCTTCTTCACCGA
PUB21-C39A-Fw	CCGAACCACTTCCGGGCCCCGATATCTCTCGAC
LYK5-qF	ACAACAACTTGCCTAACTGCG
LYK5-qR	TACTGTGAGCCACTGCATCG
FLS2-qF	TGGGGAGATCGTAGCAGTGA
FLS2-qR	CAGCGTCCACCTTGAGTGAT
WRKY22-qF	GTGGAGCGAAACAGATCCGA
WRKY22-qR	GGGTTTGTTTGAGTCGCTGC
RbohF-qF	AAGCAGTGGCAAAAGCGAAG
RbohF-qR	TCAAAGCTCGAGCCGCTAAA
EARLI1-qF	TGTCACTATTGGCACACCCC
EARLI1-qR	GGCTGAGTGAAAGCGGGATA
LOX2-qF	GAGGAAGCTCGAACACACGA
LOX2-qR	GAGCTTTGGCTAGCCTCCAA
LOXA-qF	GTGACGAGGTCAACAGGGTT
LOXA-qR	GCAATGACGAATGGCTCGAC
JAZ-qF	ATGTCGAGCTCAGGCCAAAA
JAZ-qR	TGCTCCGTTCTGGCTTCATT
GGPPS7-qF	AGCGTGTGCGTCCTCTTTTA
GGPPS7-qR	CTTTGACCGTCTCCGCTCAT
HPT-qF	TCGTCCCTACACTTGGGCTA

HPT-qR	GAAGCGAGAGGAAGATGGGG
KAH-qF	TTTGGTAGCCCGTGTGTGAT
KAH-qR	ATGCCCCTTGAGTGGACTTG
CHS2-qF	CACTGTTCTTCGCCTCGCTA
CHS2-qR	AGCTGATACAACGGACGCTC
FLS-qF	CCTTGTGGAGAGCCCTTGTT
FLS-qR	ACCTGTTTGCTGGTAGTCCC
PAL-qF	TCCCCTTTTCGTACATCGCC
PAL-qR	ACGCAAGTCCCTCTTTAGGC

ChIP-qPCR

Primers names	Primer Sequence (5'-3')
CsPUB21pro-I-Fw	AGGAGGAAATATAACTCAAA
CsPUB21pro-I-Rv	GAATAAATCAAATCAAAGCT
CsPUB21pro-II-Fw	AGGATCAGGCTACGGTGTAA
CsPUB21pro-II-Rv	CACACAATAAATTACGGGGG
CsPUB21pro-III-Fw	GTGACAGTAAATTCAACTCATC
CsPUB21pro-III-Rv	GCTGAGTTCATTAGTTTCAC
CsPUB21pro-IV-Fw	CAGTCTAATATGAGCGCCTC
CsPUB21pro-IV-Rv	GTGCTTTGAGCGTGATTTAA
CsPUB21pro-CK-Fw	GTCAAGATTAGCATACACACT
CsPUB21pro-Ck-Rv	TATCCAATGGATATACGTGT
MpPUB21pro-I-Fw	GTCAAGAACGGCACACGCAC
MpPUB21pro-I-Rv	GTTATTCATTCATTCCGATG
MpPUB21pro-II-Fw	GTGGTCTAATATGAACGCCT
MpPUB21pro-II-Rv	GTCGAAGAATCTTCTGCCAT
MpPUB21pro-III-Fw	CAACCCGTCTAGACAAATAT
MpPUB21pro-III-Rv	AAGAAAGCAAGGCCGTCCAT
MpPUB21pro-IV-Fw	ATGGACGGCCTTGCTTTCTT
MpPUB21pro-IV-Rv	CAGTTTACTTGACTGAGTG
MpPUB21pro-V-Fw	TTCCCCTCTCCTCTTTTCTT
MpPUB21pro-V-Rv	GGTCTGGCTGACTTGGCTTA
MpPUB21pro-CK-Fw	GCACACACCGATATACCAAC
MpPUB21pro-CK-Rv	ATAGAAGGAGAACTGGTGGG
BkPUB21pro-I-Fw	ATTCTCTCTTCACCCTAAA
BkPUB21pro-I-Rv	CAGGGAATGTTAATTTGCAGAGG
BkPUB21pro-II-Fw	CCTCTGCAAATTAACATTCCCTG
BkPUB21pro-II-Rv	GTCAAAAGTACATACTCTTG
BkPUB21pro-III-Fw	AATCTCCTGTGTTCCTTCCA
BkPUB21pro-III-Rv	ATCGTAGTGCTTTCAGCAT
BkPUB21pro-IV-Fw	CAGTACGCGTTTAGCAGTCA

BkPUB21pro-IV-Rv	CTTGACGGAGTTGCTCATGC
BkPUB21pro-CK-Fw	GCACACACCTACATACCTAA
BkPUB21pro-CK-Rv	TAGAAGGAGCATTGGTGGGT

EMSA probes

Primers names	Primer Sequence (5'-3')
PUB21pro-I-Fw	AGGAGGAAATATAACTCAAA
PUB21pro-I-Rv	GAATAAATCAAATCAAAGCT
PUB21pro-II-Fw	AGGATCAGGCTACGGTGTAA
PUB21pro-II-Rv	CACACAATAAATTACGGGGG
PUB21pro-III-Fw	GTGACAGTAAATTCAACTCATC
PUB21pro-III-Rv	GCTGAGTTCATTAGTTTCAC
PUB21pro-H1H2H3-Fw	CCGTACTACAGTTGCATCCAGCTGTTGGATGCACTTGGATTAGTG
PUB21pro-H1H2H3-Rv	CACTAATCCAAGTGCATCCAACAGCTGGATGCAACTGTAGTACG
PUB21pro-mH1H2H3-Fw	CCGTACTATTGTAACATCCAGCTGTTGGATGCACTTGGATTAGTG
PUB21pro-mH1H2H3-Rv	CACTAATCCAAGTGCATCCAACAGCTGGATGTTACAATAGTACGG
PUB21pro-mH1mH2H3-Fw	CCGTACTATTGTAACATCTTGCAATTGGATGCACTTGGATTAGTG
PUB21pro-mH1mH2H3-Rv	CACTAATCCAAGTGCATCCAATTGCAAGATGTTACAATAGTACGG
PUB21pro-mH1H2mH3-Fw	CCGTACTATTGTAACATCCAGCTGTTGGATGTTCTAAGATTAGTG
PUB21pro-mH1H2mH3-Rv	CACTAATCTTAGAACATCCAACAGCTGGATGTTACAATAGTACGG
PUB21pro-mH1mH2mH3-Fw	CCGTACTATTGTAACATCTTGCAATTGGATGTTCTAAGATTAGTG
PUB21pro-mH1mH2mH3-Rv	CACTAATCTTAGAACATCCAATTGCAAGATGTTACAATAGTACGG
PUB21pro-H1H2mH3-Fw	CCGTACTACAGTTGCATCTTGCAATTGGATGCACTTGGATTAGTG
PUB21pro-H1H2mH3-Rv	CACTAATCCAAGTGCATCCAATTGCAAGATGCAACTGTAGTACGG
PUB21pro-mH4H5-Fw	GCTTGATGTTACAATAACACAGTACCACAGTTGCACCAT
PUB21pro-mH4H5-Rv	ATGGTGCAACTGTGGTACTGTGTTATTGTAACATCAAGC
PUB21pro-H4mH5-Fw	GCTTGATGCAACTGTAACACAGTACCATTGTAACACCAT
PUB21pro-H4mH5-Rv	ATGGTGTTACAATGGTACTGTGTTACAGTTGCATCAAGC
PUB21pro-H4H5-Fw	GCTTGATGCAACTGTAACACAGTACCACAGTTGCACCAT
PUB21pro-H4H5-Rv	ATGGTGCAACTGTGGTACTGTGTTACAGTTGCATCAAGC

Data S1. Putative E3 genes in the *Citrus sinensis*.

Data S2. The whole sequence of TE_00000031#DNA/*Helitron* in the genome of *Citrus sinensis*.

Data S3. DNA sequences of *PUB21* promoters.

Data S4. Antimicrobial peptides form human gut microbiome.

Data S5. Potential PUB21-binding peptides.

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